Flexibility in the Belgian electricity system

Kristof De Vos

Course Athens KU Leuven – 19/03/2026

Structure

- **Introduction**
- Challenges of the energy transition
- Balancing a renewable energy system
- Flexibility needs of the system
- Summer outlook

Kristof De Vos

Economist with a PhD in Electrical Engineering specialized in electricity market design

E-mail : kristof.devos@elia.be

2008-2015

- Master in commercial engineering (KU Leuven)
- PhD in electrical engineering (KU Leuven)
- Postdoc in electricity market research
  - Design of electricity markets
  - Integration of wind power in electricity markets
  - Impact of wind power on sizing and allocation of balancing reserves

2015-2019

- System services manager at Elia on **Adequacy**
  - Design of adequacy and capacity remuneration mechanisms
  - Product Manager of Belgian Strategic Reserves

2017-2023

- System services manager at Elia on **Balancing**
  - Dimensioning of Elia's reserve requirements
  - Determination of flexibility needs for renewable energy systems
  - Design of balancing products and markets
  - System integration of offshore wind power

2023-…

- Manager **Flexibility** studies

Elia Group in Europe

REGULATED ACTIVITIES

Develop & maintain the grid

System operations & balancing

Market facilitation

Renewables integration

50hertz

- Northeastern Germany TSO operator
- On- and offshore transmission systems
- 80% owned by Elia Group (20% KfW)
- Monopolistic position in Northeast Germany

elia

- National TSO
- On- and offshore transmission systems
- 99.99% owned by Elia Group
- Monopolistic position in Belgium

nemolink

- 50/50 JV between Elia and National Grid (UK)
- Grid interconnection between BE and UK
- 50% owned by Elia Group

elia group

19,276 km of HV infrastructure

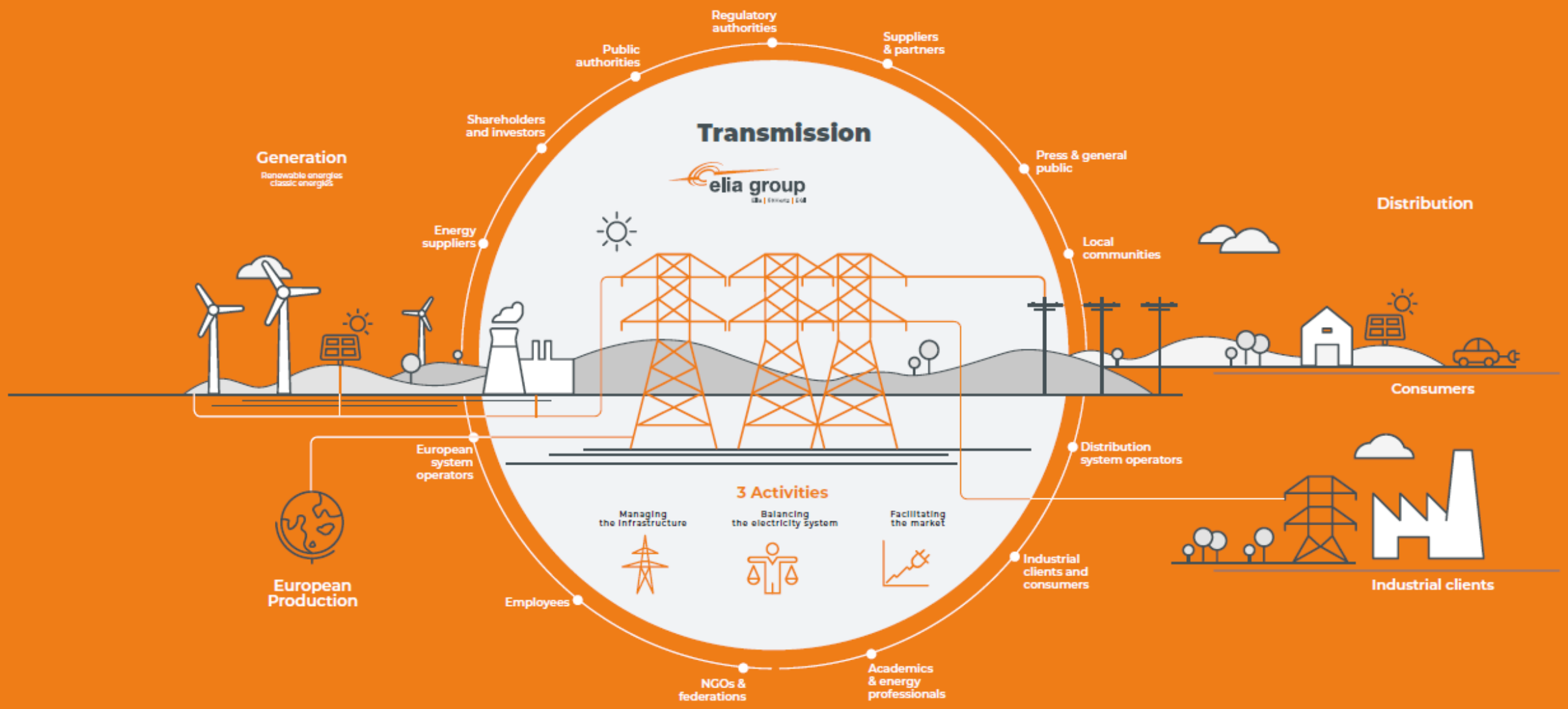
€ 18.5 billion Regulated Asset Base

30 Million End Users

99,9% Grid Reliability

Top 5 TSO in Europe

We connect distribution and generation



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Electricity grids as key European & national asset

- Feeding industry, households, airports, ports, etc. but also hospitals, elevators, traffic lights,…
- Impact on society well witnessed during recent black-out in Spain & Portugal (and CZ);
- Economic dependency well witnessed during (and still ongoing) dependency of beyond EU-regimes to fuel our economy;
- **Grids in focus at European level**: EU Action plan for grids, Draghi-report; affordability action plan, Grids Package
- **Grids in focus at National level**: capacity mechanisms to support generation, tariff-design, exemptions for industry, competitiveness;

Making the Green Deal a reality: the road to decarbonisation

elia | Elia Group

Higher level of ambition for energy and climate goals

More attention to strategic autonomy

European Green Deal

July 2021

Fit for 55 package

Series of landmark energy and climate initiatives to make the Green Deal objectives concrete. Goals to reduce emissions by 55% by 2030 and reach climate neutrality by 2050.

May 2022

REPowerEU package

Increased ambition for energy and climate targets to address the energy crisis and the impact of the Russian invasion of Ukraine.

March 2023

Net Zero Industry Act

Technologies needed for the Green Deal should be produced in the EU. The Act is a response to similar measures enshrined in the US Inflation Reduction Act.

"We have since long argued that the fight against climate change is a must. A must for our planet, a must for our economic prosperity, and a must for our ***strategic independence****."*

Ursula von der Leyen, 1 February 2023

2030 Targets

Decarbonisation

55% *less GHG emissions by 2030* compared to 1990 levels

Energy Efficiency

39% *Reduction in primary energy consumption by 2030*

Renewable electricity

65% *of total electricity production by 2030**

Offshore wind

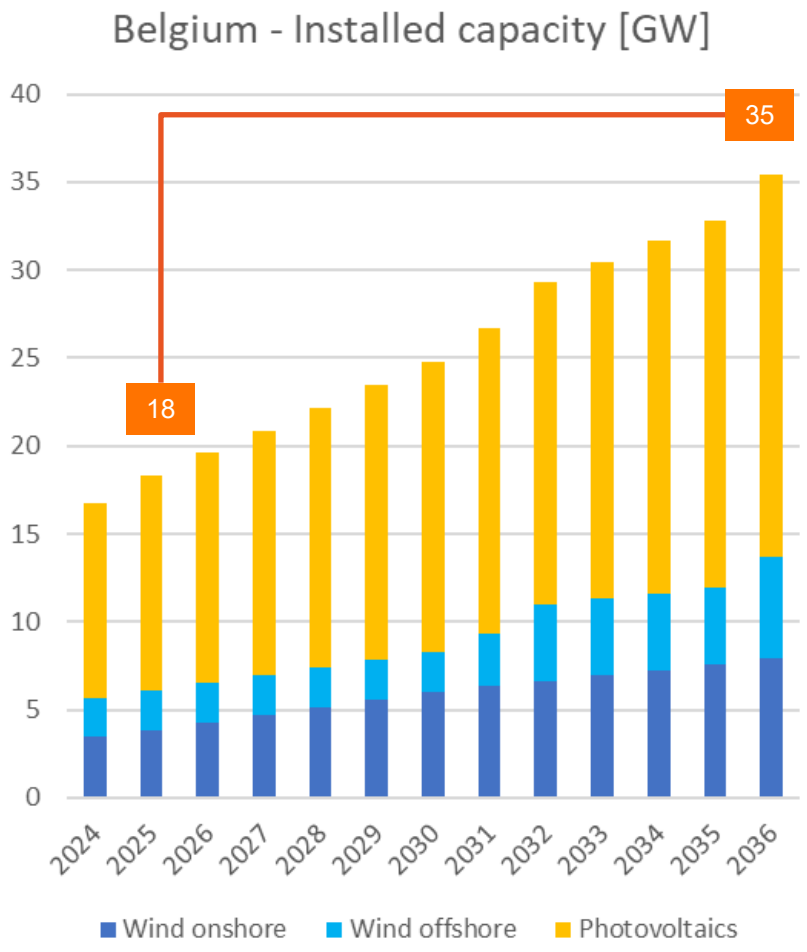
60GW *by 2030 & towards 300GW in 2050, compared to 12GW in 2020.*

**European Commission estimation bason on binding GHG targets*

Projections on installed renewable capacity in Belgium

Projections on installed renewable capacity is based on Adequacy and Flexibility study 2025 – Current Commitment Scenario :

- **Offshore generation of 2.3 GW** in 2030 (further increasing after)
- **Photovoltaic capacities almost 17 GW** in 2030
- **Onshore wind up to 6 GW** in 2030

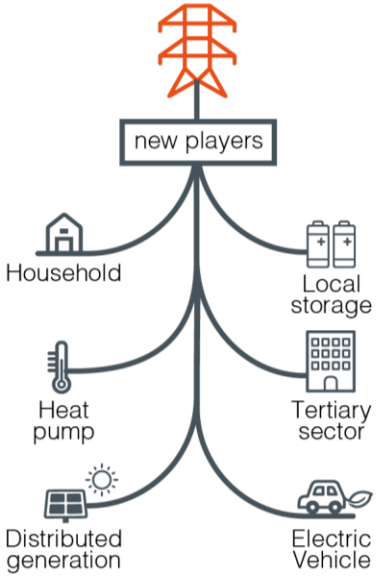


Variability and limited predictability of these resources are expected to result in pressure on reserve capacity requirements and system operation costs in general

Major trends in the last decade

The development of variable generation

Decentralisation, digitalisation and new players



The regionalisation of the electricity sector

Consequences

→ More and longer distance transmission & interconnection

→ Intensified coordination local (DSO-TSO) & supranational (ENTSO-E, CORESO)

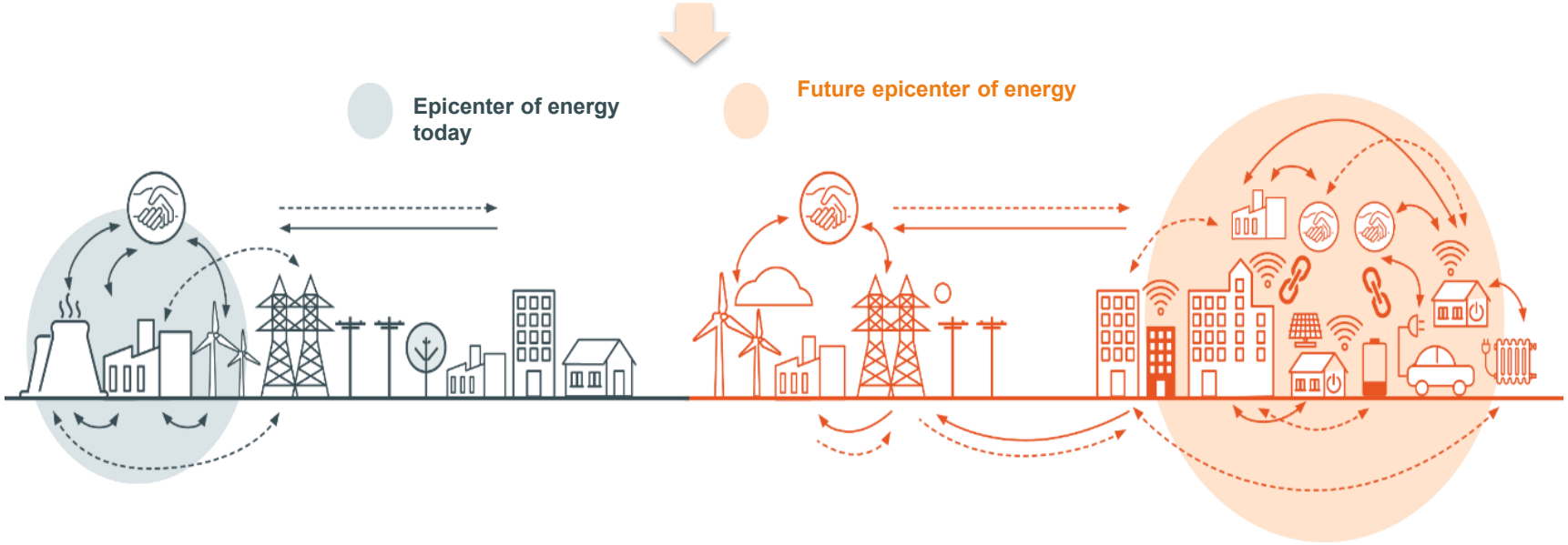
FLEXIBILITY

→ **More flexibility needs to balance a system** with high amounts of variable renewable generation

→ **More flexibility available at end-user level** and appearance of new business models

Electrification and the energy transition

- Renewable energy will increase both in the overall energy mix and in the electricity mix
- The share occupied by electricity in final consumption will increase with electrification
- Digitalization and market design will facilitate an active participation of consumption

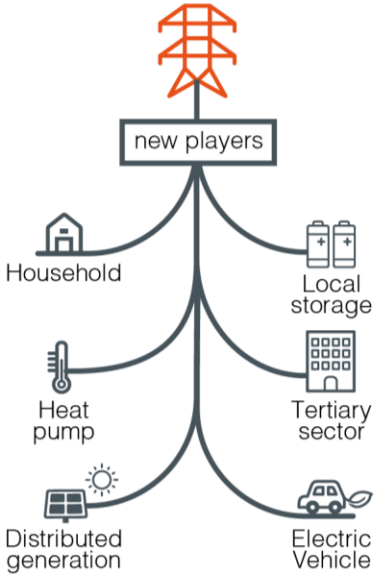


Generation follows consumption → **Demand will follow generation**

Major trends today

The development of variable generation

Decentralisation, digitalisation and new players



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Electrification of energy use

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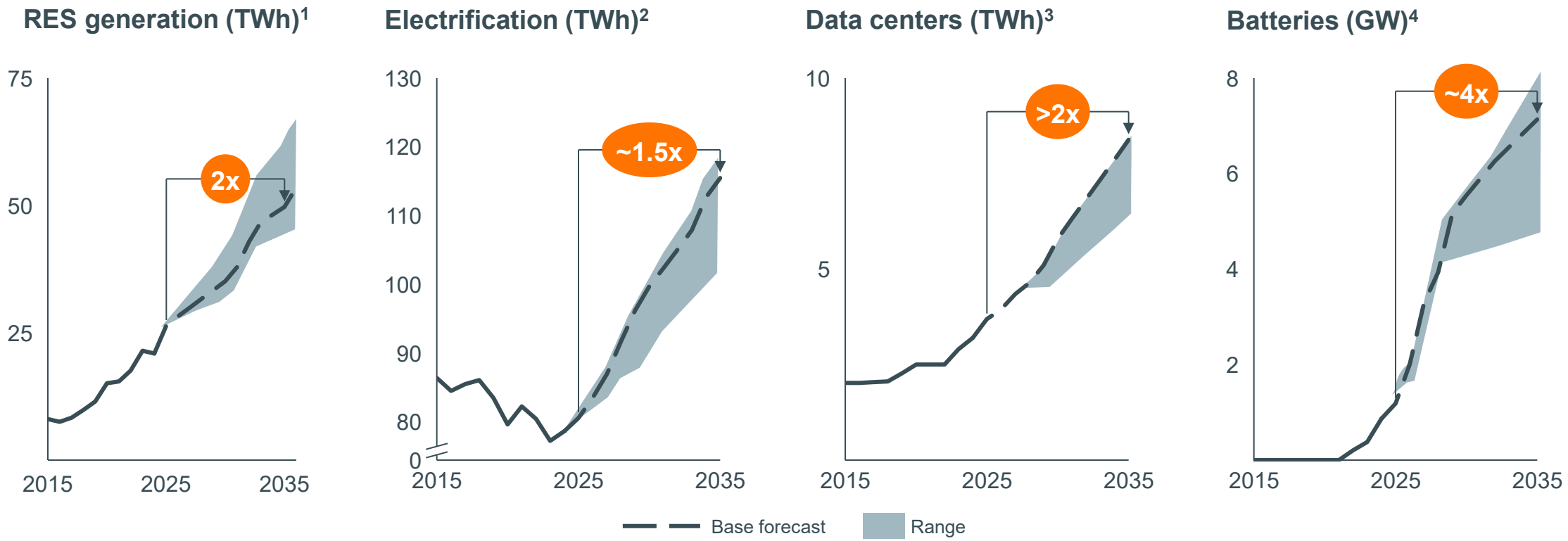
CONGESTION

→ **Increasing attention to managing scarce network capacity**

Approved plans anticipated unprecedented levels of new generation, electrification, data centers, and storage capacities in Belgium

Projections based on societal choices as outlined in current plans



1. Includes Solar PV, onshore and offshore wind; Load factors: Solar PV 12%, Onshore wind 22%, Offshore wind 38% 2. Total electricity demand excluding data centers load
3. 2015-2019 load based on installed capacity 4. Both large- and small-scale battery storage
Note: Base forecast reflects Current Commitments scenario; Range built on Constrained Transition scenario (lower bound) and Prosumer Power (upper bound)
Source: Elia Adequacy and Flexibility Study (2024-2034, 2026-2036); Elia Open Data Portal

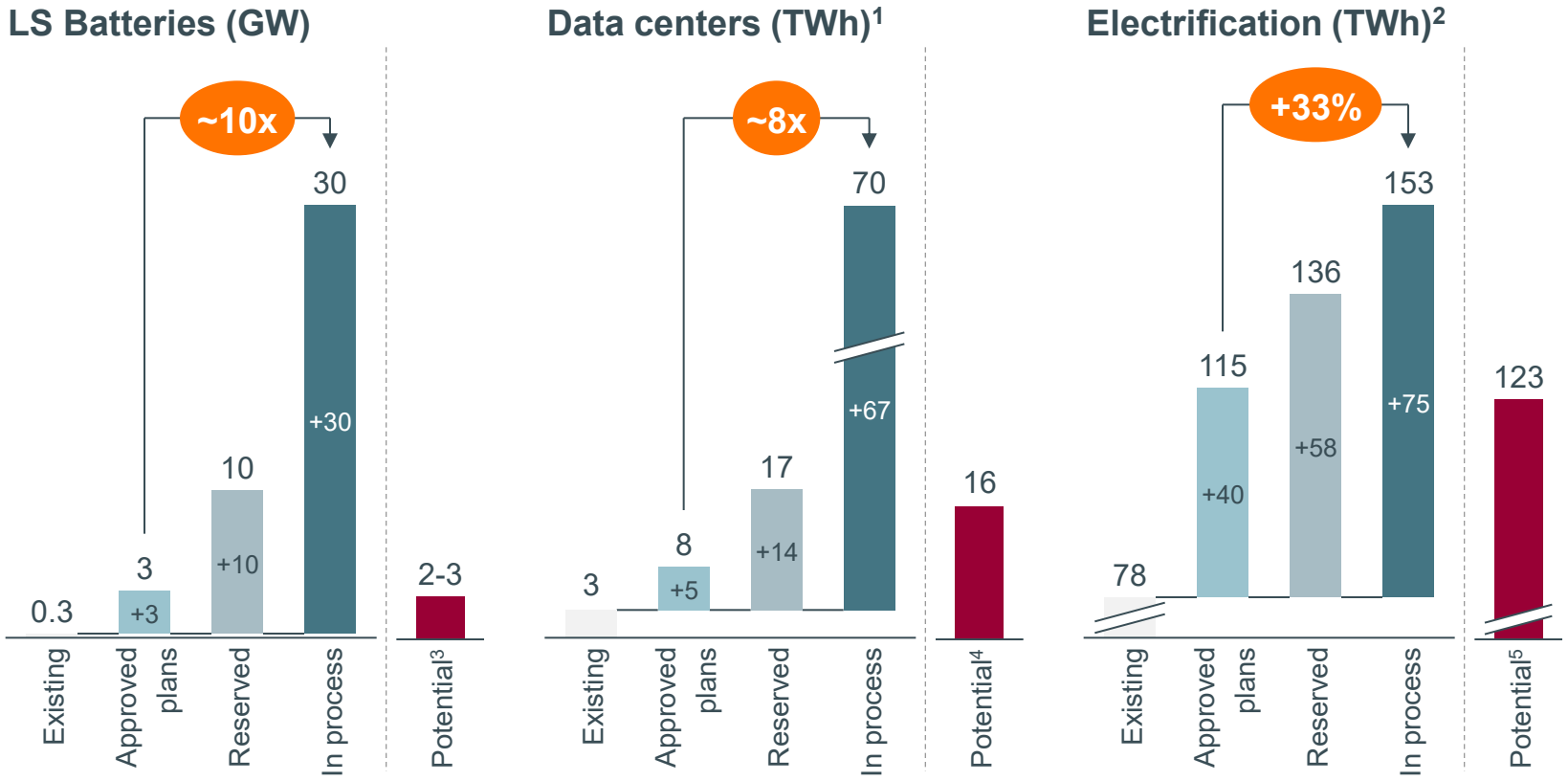
Unprecedent changes in the amount of requests to connect to the grid

Note: as of mid 2025

Surge in new assets
Data centers & batteries scaling much faster than expected

Speculative behavior
Anticipated or speculative projects inflating the queue

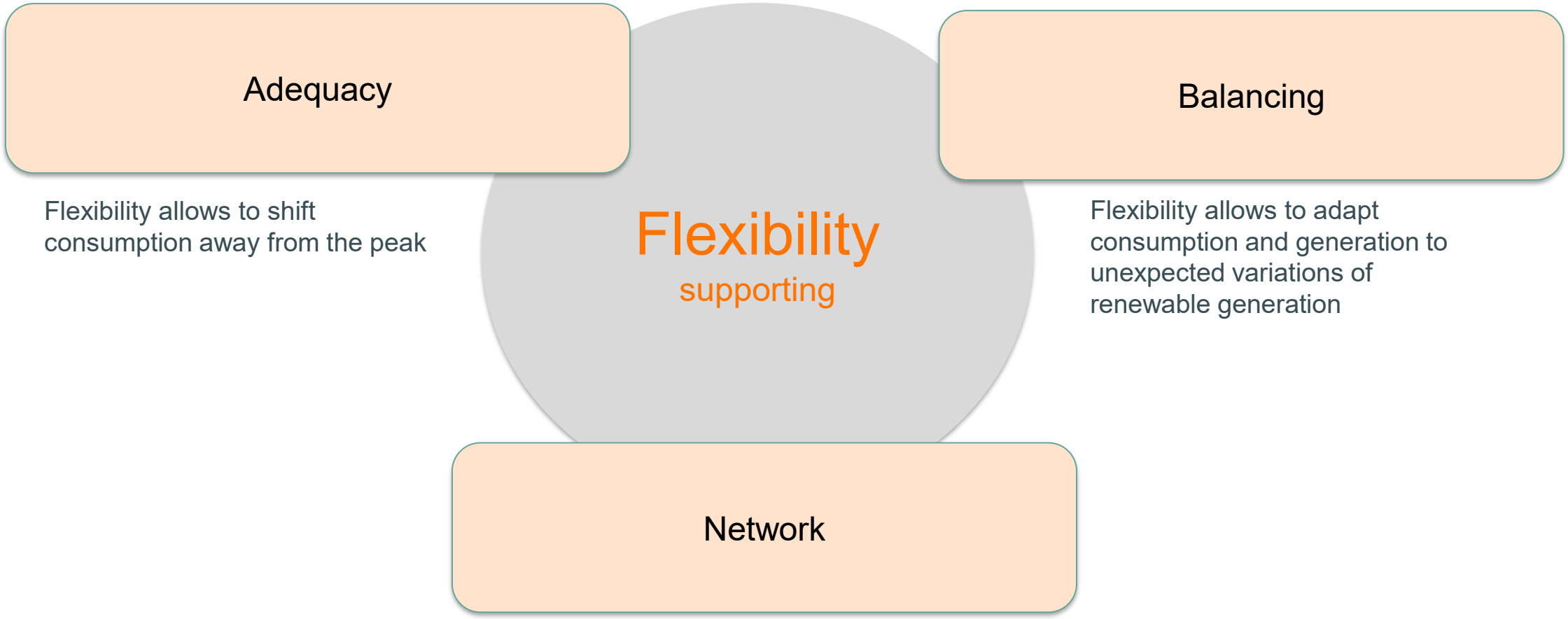
Electrification progress
Reservations more than on track with approved plans



On top of this, there's a quick rise in requests on the distribution grid from SMEs

1. Converted from GW to TWh using 85% load factor 2. Electrification figures based on total electricity demand minus data centers
3. EnergyVille 2040 value 4. From BCG The Power to Compute 5. From AdeqFlex (2026-2036), in "Current Commitments" scenario
Note: "Approved plans" refers to AdeqFlex (2024-2034) Study; "Reserved" coming from Elia's hosting capacity map; "In process" adding orientation studies on top of reserved
Source: Elia Adequacy and Flexibility Study (2026-2036); EnergyVille Paths 2050 (April 2025); Elia Transmission Belgium; Boston Consulting Group The Power to Compute (May 2025)

Flexibility in the centre of the energy transition



Flexibility allows to shift consumption away from the peak

Flexibility allows to adapt consumption and generation to unexpected variations of renewable generation

Flexibility allows to desaturate grid congestions and optimize grid investments

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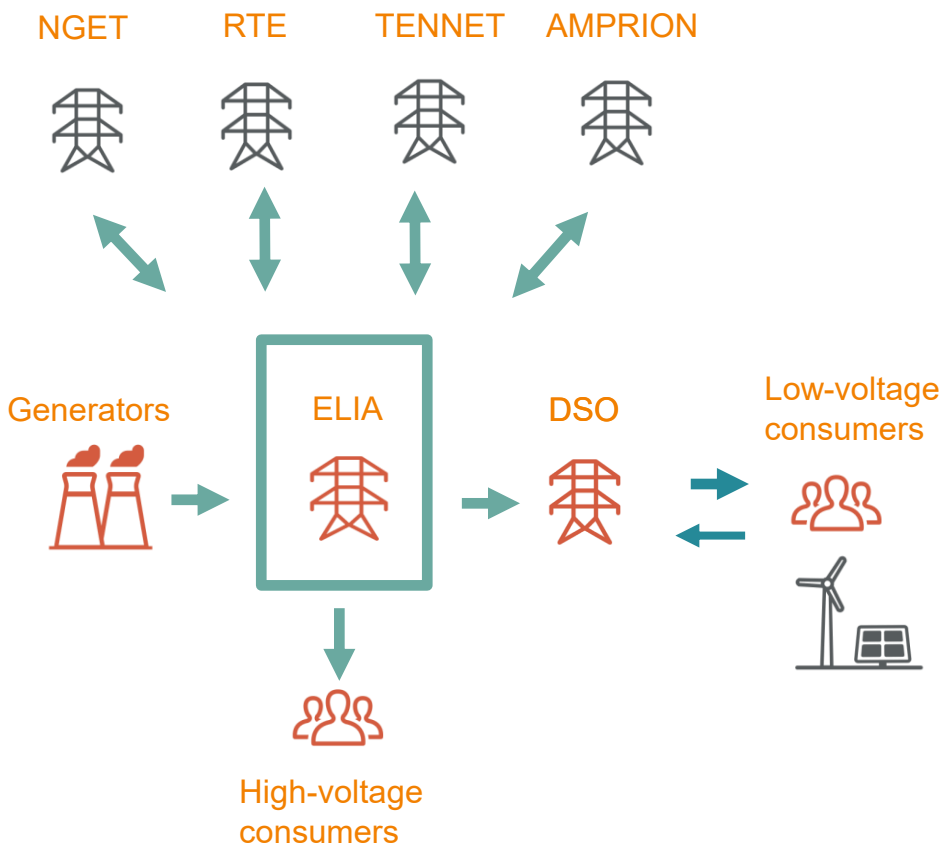
Physical transport of electricity



- Large part of electricity is generated by market parties in power plants (including offshore wind parks)
- Electricity is transformed to high voltage and transported by the TSO to large consumers and other regions
  - Nationally by Elia : 70 kV to 380 kV
  - AC interconnections to France and The Netherlands
  - HVDC Interconnections to Great-Britain and Germany
- Electricity is transformed to lower voltage and distributed by the DSO to end-consumers (e.g. Fluvius, ORES). Electricity is used in households, industry and tertiary sector
- Note that the presence of decentralized generation (e.g. photovoltaics and combined heat and power) this flow is becoming become bi-directional

Facilitated via an electricity market

- **Wholesale markets**
  - Via forward, day-ahead and intra-day markets
  - Via bilateral exchanges and power exchange (e.g. EPEX / NORDPOOL)
  - Via explicit capacity or implicit via European flow-based market coupling
    - Access to foreign generation / consumption
    - Constrained by interconnection capacity

- **Retail market**
  - Suppliers deliver energy to end-consumer
    - Concludes a contract with end-consumer
    - Invoices energy consumption (and other invoice components)

Balance management

Central role of the market participant

- Responsibility of TSOs is to maintain stable frequency levels and avoid black outs
- This responsibility is pushed as much as possible to the market, i.e. BRPs which need to balance their individual portfolios

MARKET PARTIES

Day-ahead market

- Each market party nominates its portfolio per hour based on predictions
- This balance is achieved by means of using various flexibility sources, by means of contracts or own flexibility means

Intra-day market

- Restoring the balance based on new forecasts :
  - by means of intra-day market (until 1 hour before real-time), or
  - through own flexibility means

Intra-day to real-time

- Imbalance price of Elia gives incentive to restore balance
- Additional deviations (forced outage, wind power variations) can be dealt with bilaterally or by means of own flexibility means

ELIA

Real-time balancing

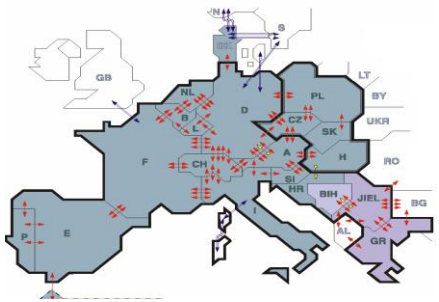
- Elia manages the residual aggregated system imbalance
  - with imbalance netting with other regions
  - With balancing services (aFRR and mFRR), or,
  - with non contracted balancing energy bids from market parties

Uncertainty

TSO Balance Management (1)

Maintaining stable frequency

Solidarity mechanism between TSOs to keep freq = 50Hz

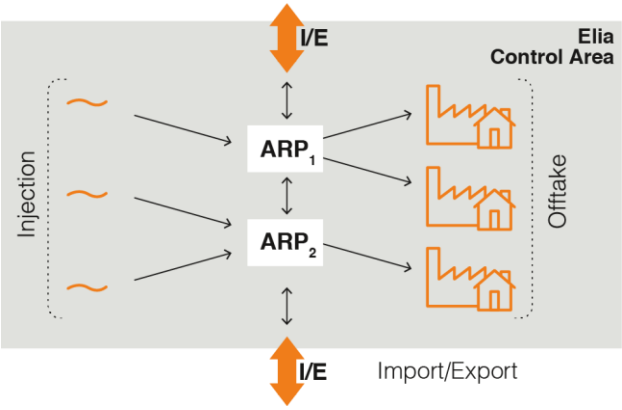


How?
Automatic activation of FCR in synchronous area upon frequency deviations

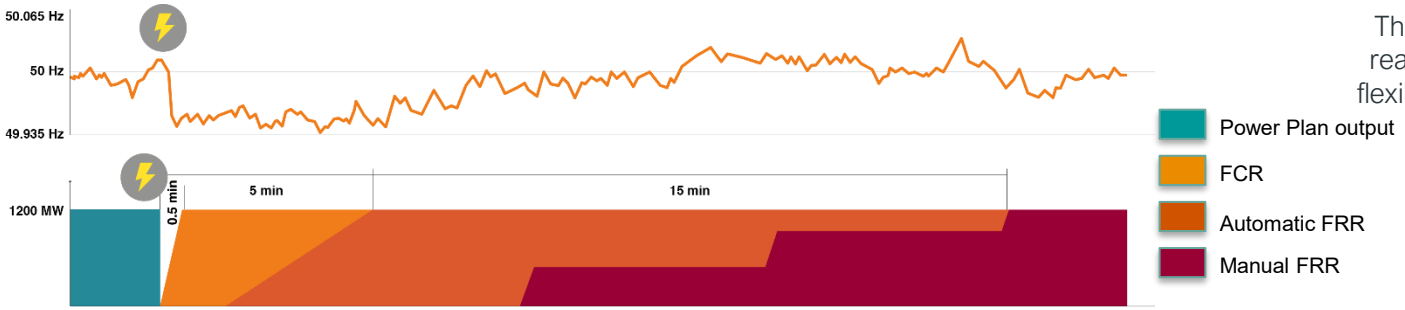
Each TSO is responsible to keep its control area (= LFC block) balanced

How?
Activation of automatic FRR (for small imbalances) and manual FRR (for bigger imbalances). To this market is referred to as the "Balancing Market"

Each Balancing Responsible Party (BRP) is responsible to keep its portfolio balanced in a 15 min basis



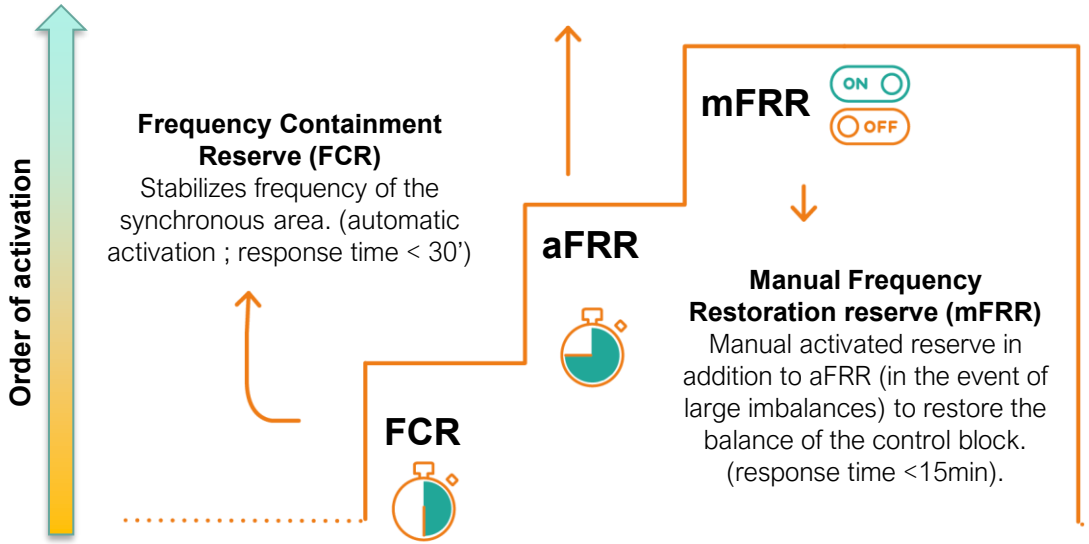
How?
Through adequate planning/forecasting and real time adjustments. BRPs use for this own flexibility as well as DA/ID trading opportunities



TSO Balance Management (3)

Balancing capacity products

All technologies All players All voltage levels



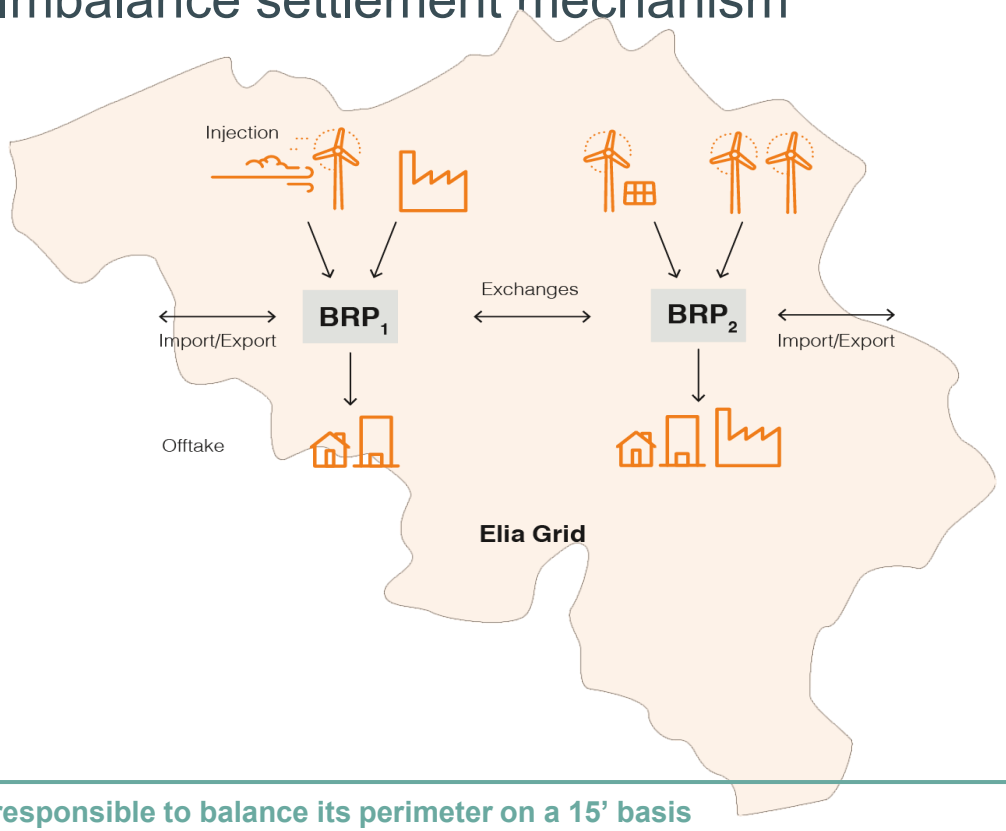
Pre-Contracted Volumes (=Balancing Capacity) availability guaranteed by contract	**"Free Bids"** Not pre-contracted; availibitliy only guaranteed during bid validity
• mFRR Standard (mFRR Flex)	• Free mFRR Bids
• aFRR Up • aFRR Down	• Free aFRR Bids
• FCR symmetric	• N.A.

Ensure a cost-efficient reservation of capacity to cover imbalances

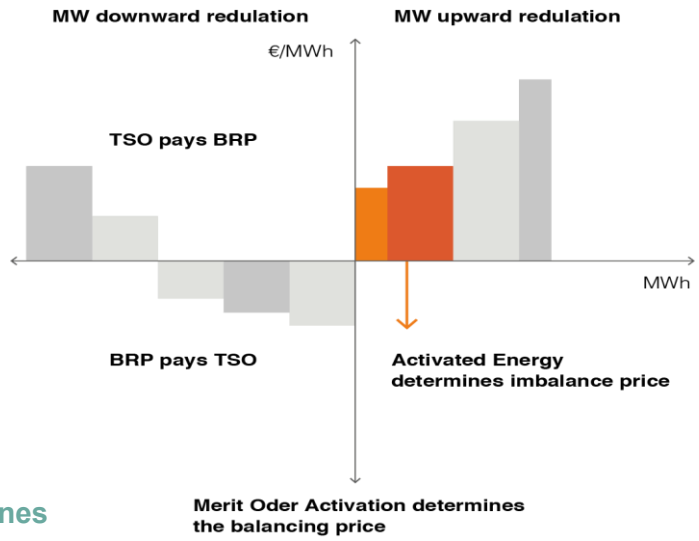
Minimize the activations costs of reserves

Relation BRP & TSO Balance Management

Imbalance settlement mechanism



Marginally activated Bid in the merit order activation determines the Imbalance Price



Residual system imbalances determines the imbalance price

BRP is responsible to balance its perimeter on a 15' basis

- BRPs nominate their positions and real-time positions must be balanced
- But BRPs can face portfolio imbalances due to prediction errors or outages

- BRP pays an imbalance price if the final position of its perimeter is not balanced
- BRPs react on real-time imbalance tariffs to restore their (and others') perimeter balance

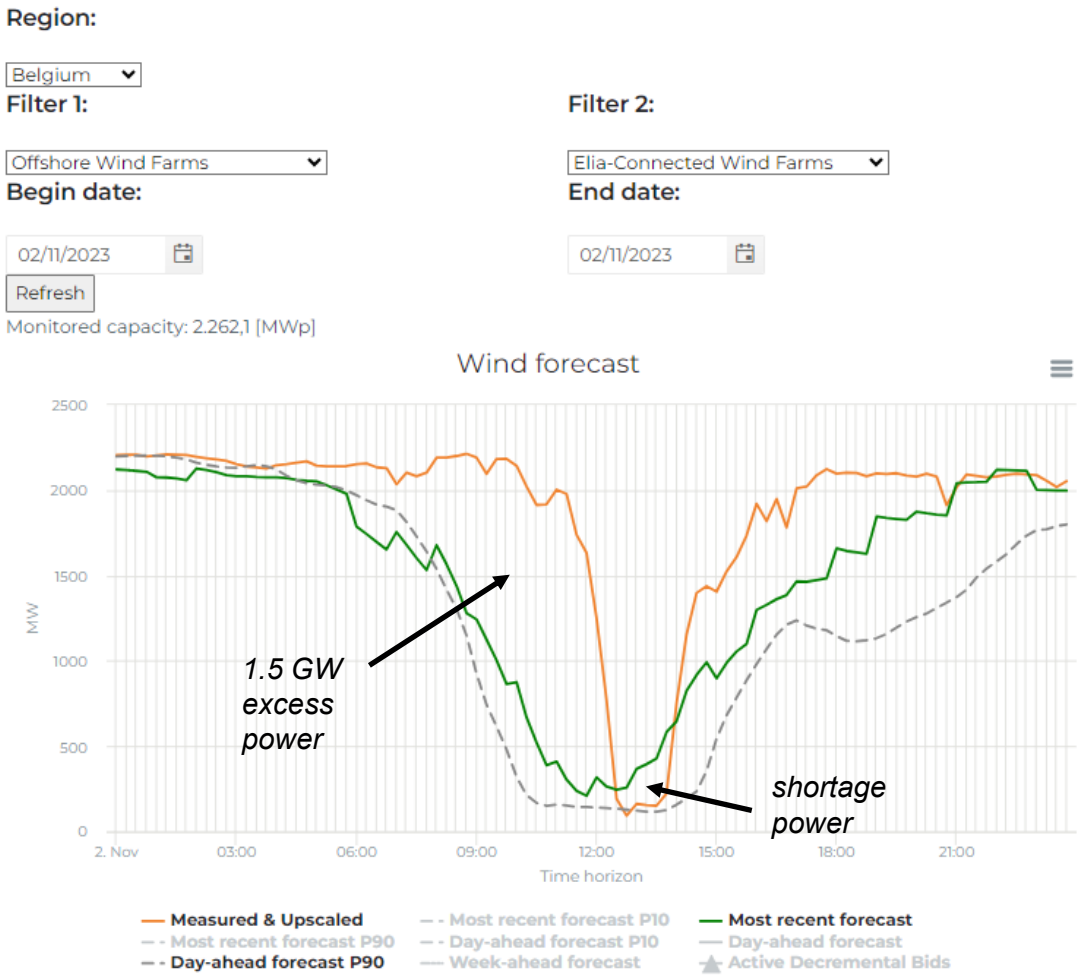
In real time, the TSO resolves residual imbalances in real-time

- By activation of Balancing Services (aFRR and mFRR)
- Balancing actions by TSOs are reflected in **imbalance prices**
  - Shortages: TSO generally pays a high price for upward reserves
  - Excess: TSO is generally pays a low (or negative) price for downward reserves

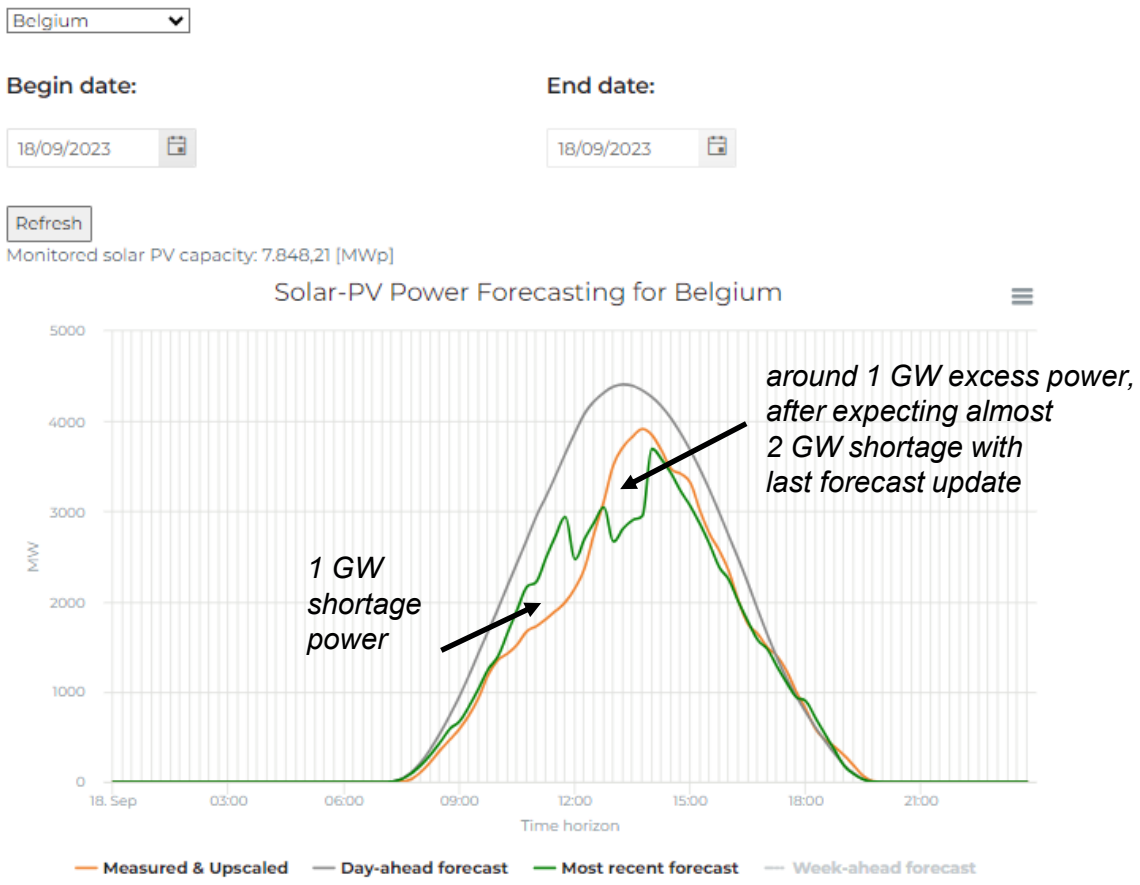
Imbalance price penalizes BRPs causing the problem

Illustration of renewable forecast errors

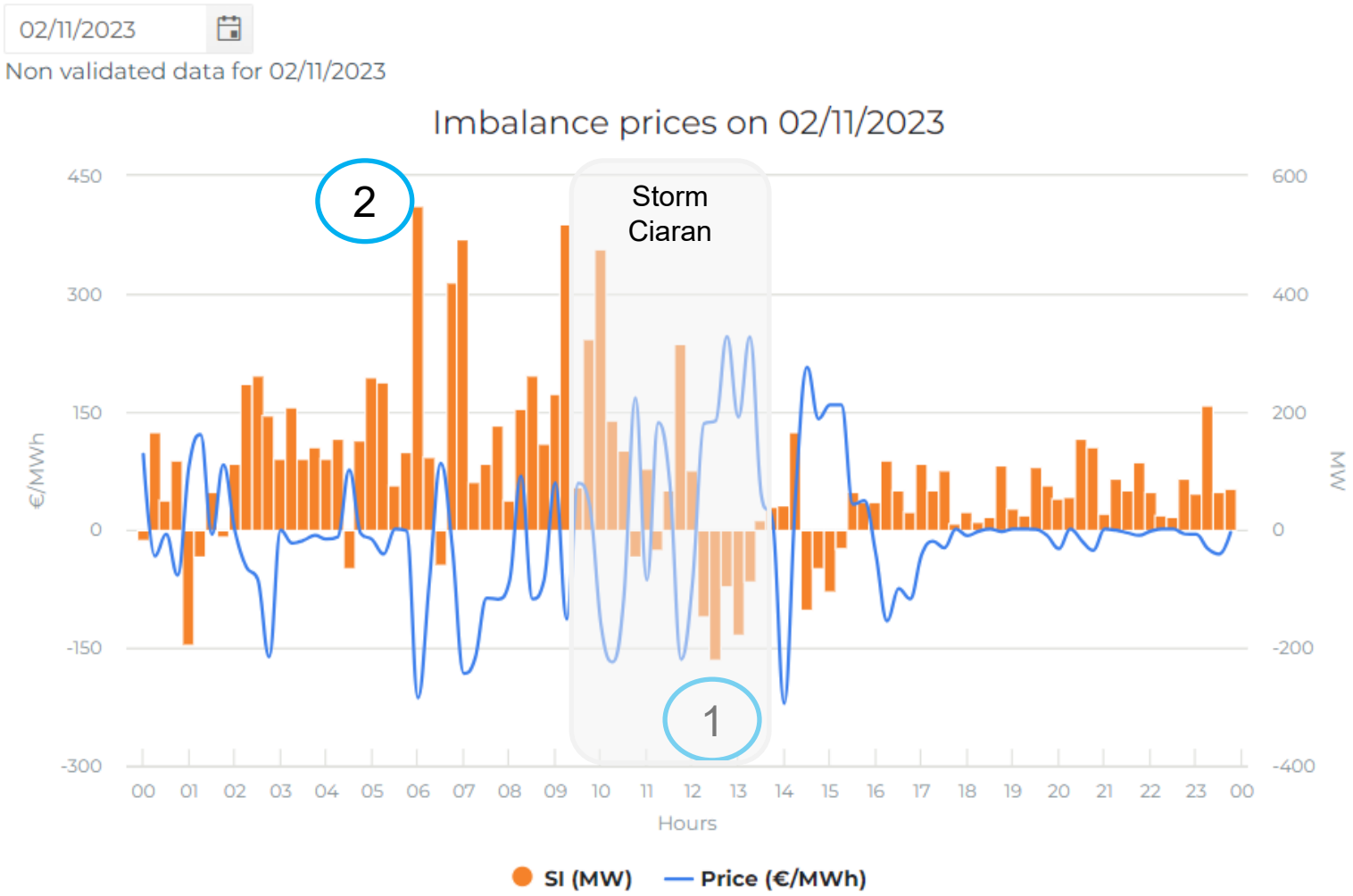
Offshore wind power during storm event – Ciaran 2/11/2023



Solar fleet behavior during a regular day – 18/09/2023



Example of Elia's imbalance prices



https://www.elia.be/en/grid-data/balancing/imbalance-prices-15-min

Snapshot of imbalance volumes and prices taken from Elia's website

1. Large shortages in the system typically result in a high price (set by the most expensive upward offer activated), this sets the price on the balancing market to be paid by the BRPs facing a shortage in their portfolio, and received by the BRPs

2. Large excesses in the system typically result in a low or even negative price (set by the most expensive downward offer activated)
   1. Generally, BRPs will pay for a downward activations (e.g. facing fuel cost savings)
   2. But most expensive sources (e.g. wind power losing certificates) want to be compensated.

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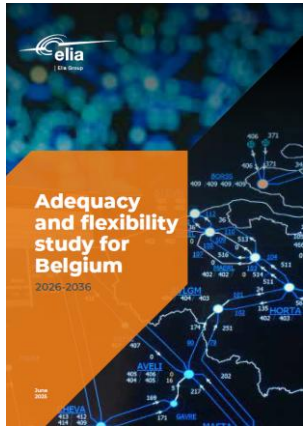
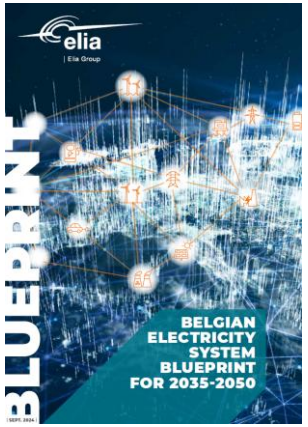
A few insights from long term simulations we perform at Elia based on state-of the art fundamental analysis of the power system

2026-2036



June 2025

2035-2050



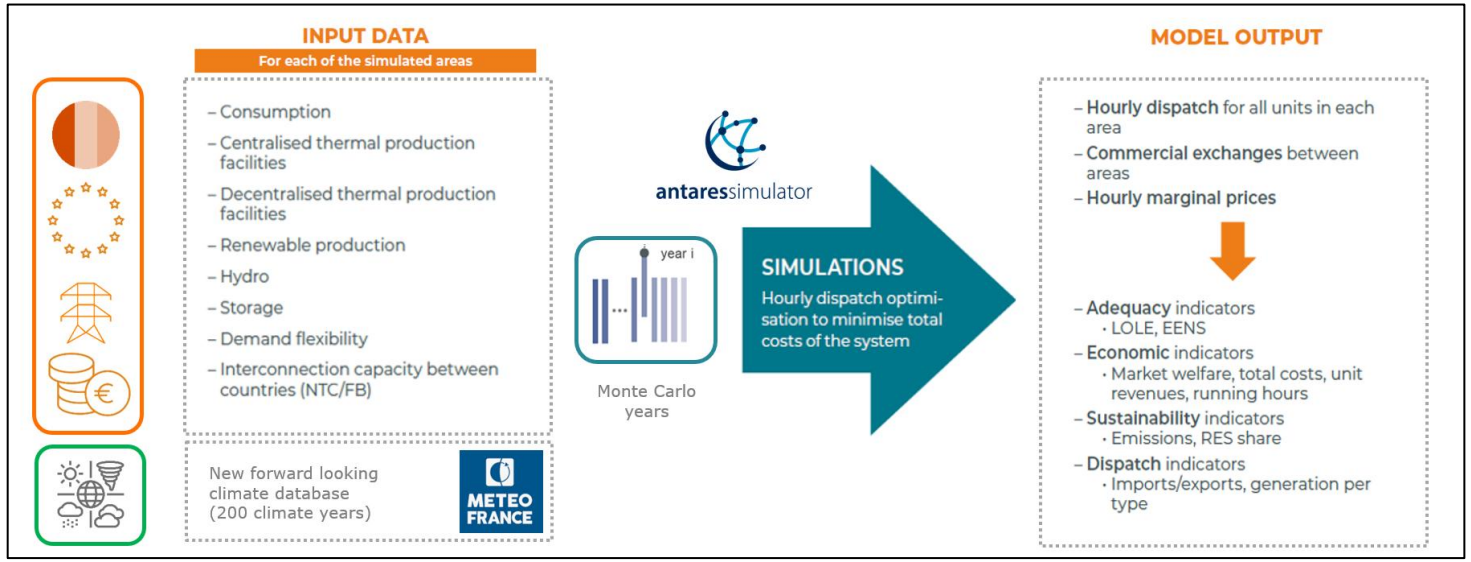
Sep 2024

2028-2038 (2050)

Upcoming

Always European scope. Current bidding zones

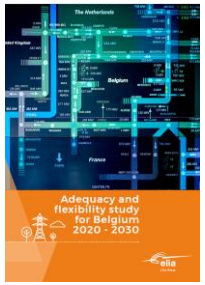
Core flow based including Advanced hybrid coupling and Min70% RAM rule.
40 Dimensions flow-based domains

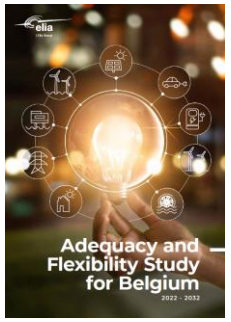


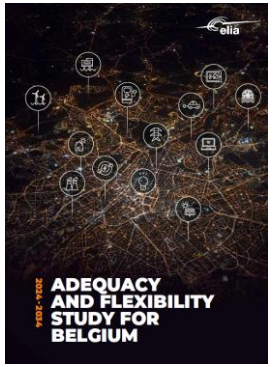
- **Hourly simulations** (8760 hours per year) with multiple **weather years** (200 forward looking weather years including climate change).
- Results are always an average of those 200 years
- Units >100 MW in Europe are individually modelled. Rest is aggregated
- Unit commitment with perfect foresight

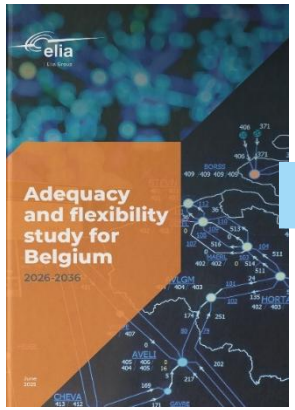
The FNA - Successor of Elia's Adeqflex'25

- Elia traditionally combined its Adequacy and Flexibility needs study in one report
- The FNA requires TSOs/DSOs to conduct analysis while the NRA or a Designated entity drafts the report
- The FNA will require a modification of the national legal framework (Article 7bid) in direction of *:
  - Elia will be appointed as designated entity for Belgium*
  - A separate Flexibility study subject in cooperation with DSOs and subject to CREG approval
  - Based on scenarios and target years of the Adequacy study









Planning :

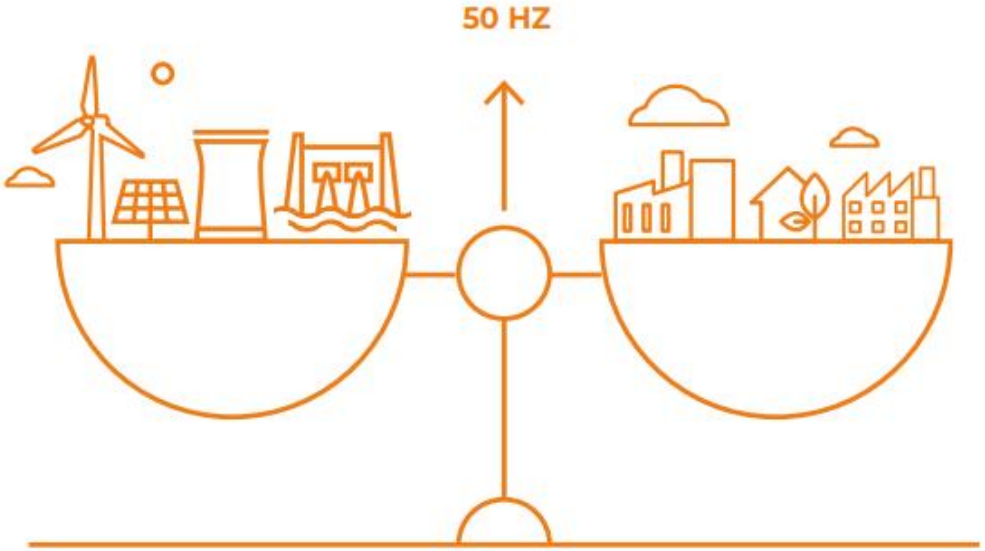
• Summer 2026	First FNA based on Adeqflex'25 scenarios
• June 2027	Elia new adequacy study
• Q2 2028	Second FNA and iterated every two years

**The Cabinet confirmed that Elia will be proposed as Designated Entity*

Flexibility : what are we talking about …

"The extent to which a power system can modify electricity production or consumption in response to variability, expected or otherwise" - International Energy Agency

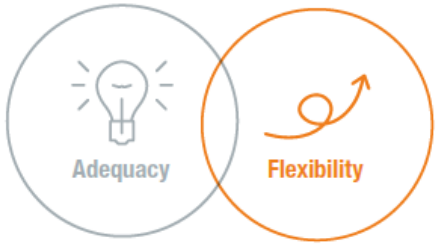
FLEXIBILITY DRIVERS
- Variability of the demand
- Variability of generation
- Forced outages



FLEXIBILITY SOURCES
- Generation units
- Demand-side assets
- Electricity storage
- Interconnectors

Adequacy study

The adequacy study investigates the required generation capacity to cover peak demand periods.



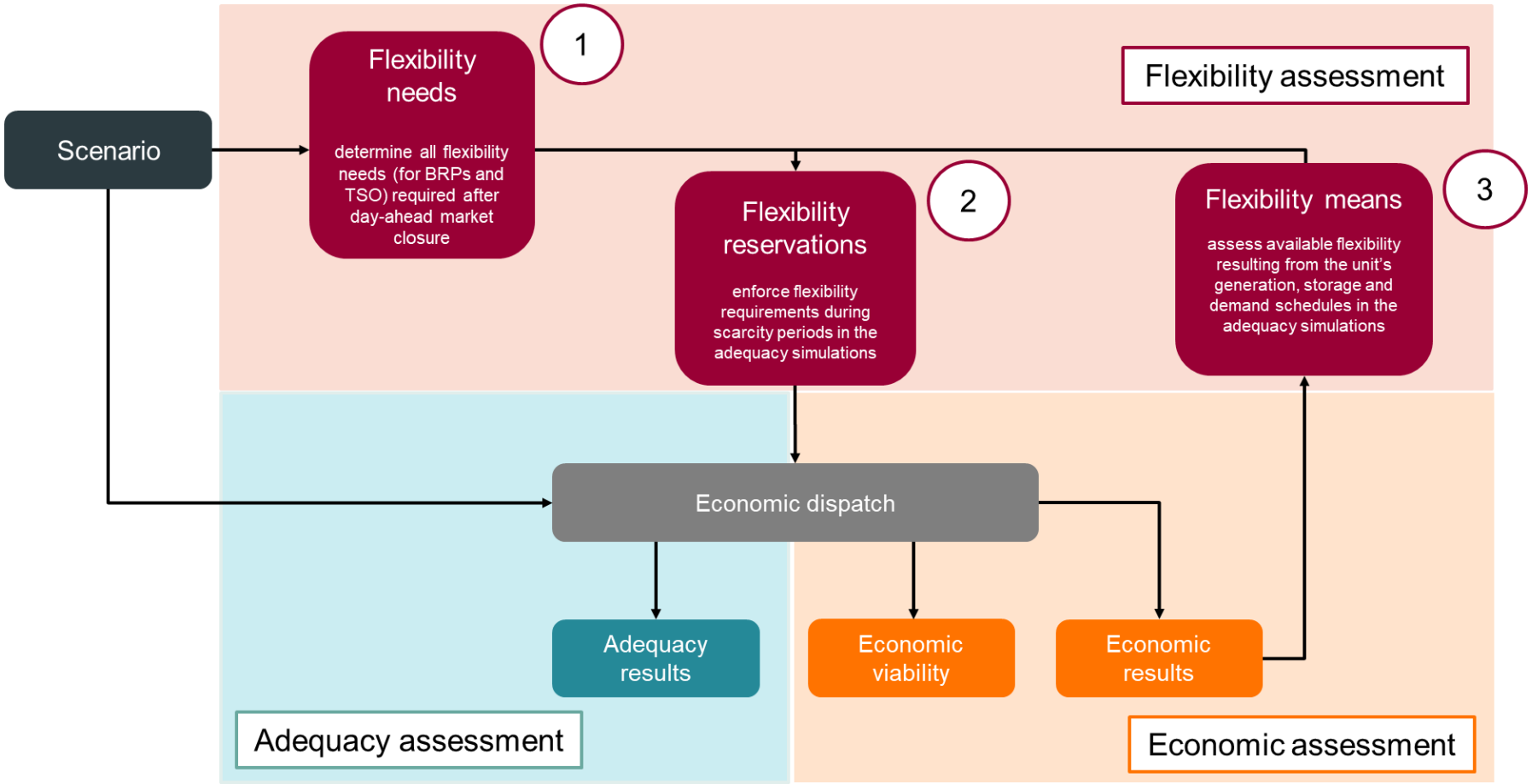
Flexibility Study

The flexibility study investigates the required technical characteristics to deal with demand and supply variations.

A focus on the ability to manage (un)expected variations

- *Elia's economic dispatch model is based on <u>perfect forecasts and hourly resolutions.</u>*

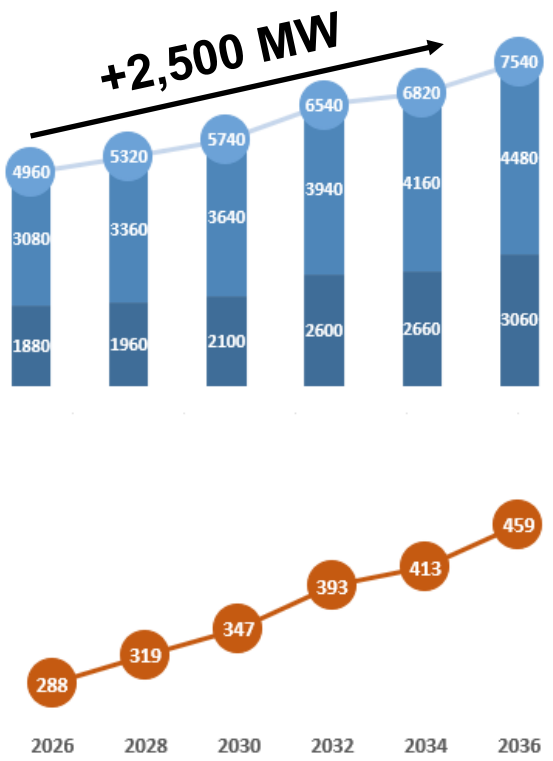
- *The short-term flexibility needs studies the capability to cover <u>fast and unexpected variations caused by forecast errors</u> of generation and demand, as well as forced outages of grid infrastructure*



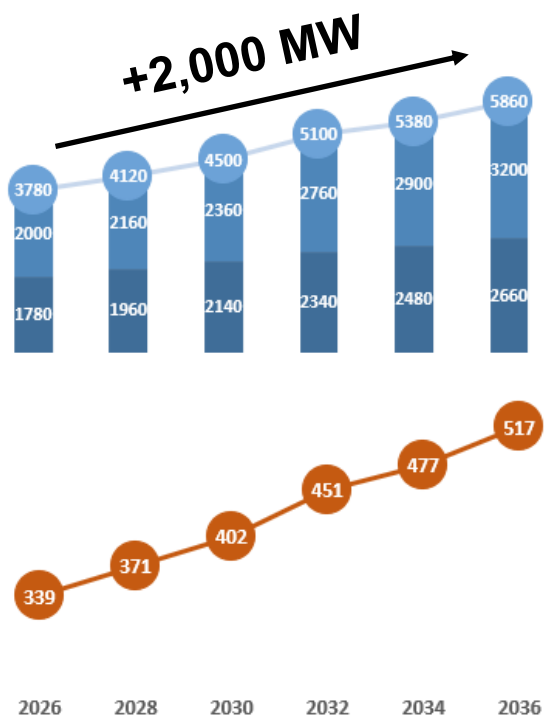
Latest analyses re-confirm increasing flexibility needs towards 2036 driven by increasing renewable capacity

- *The results represent the required capacity to manage prediction errors of renewable capacity in intra-day and real-time*
- *The flexibility needs presented have to be managed by the market (short-term markets) and Elia (reserve capacity)*
- *Without action, the share to be covered by the TSO through reserve capacity will increase proportionally*

Upward needs - MW



Downward needs - MW



Total Flex [MW/5h]

Slow Flex [MW/5h]

Fast Flex [MW/15min]

Ramping Flex [MW/5min]

By 2036, the Belgian system will require

- **6 - 7 GW of flexibility** in the last hours before real time,
- of which **3 GW needs** to be able to **react in the last quarter hours**,
- and up to **0.5 GW needs** to **react within 5 minutes**.

**Ramping flexibility is a subset of the fast flexibility needs*

The assessment of available flexibility means is to large extent based technology specifications

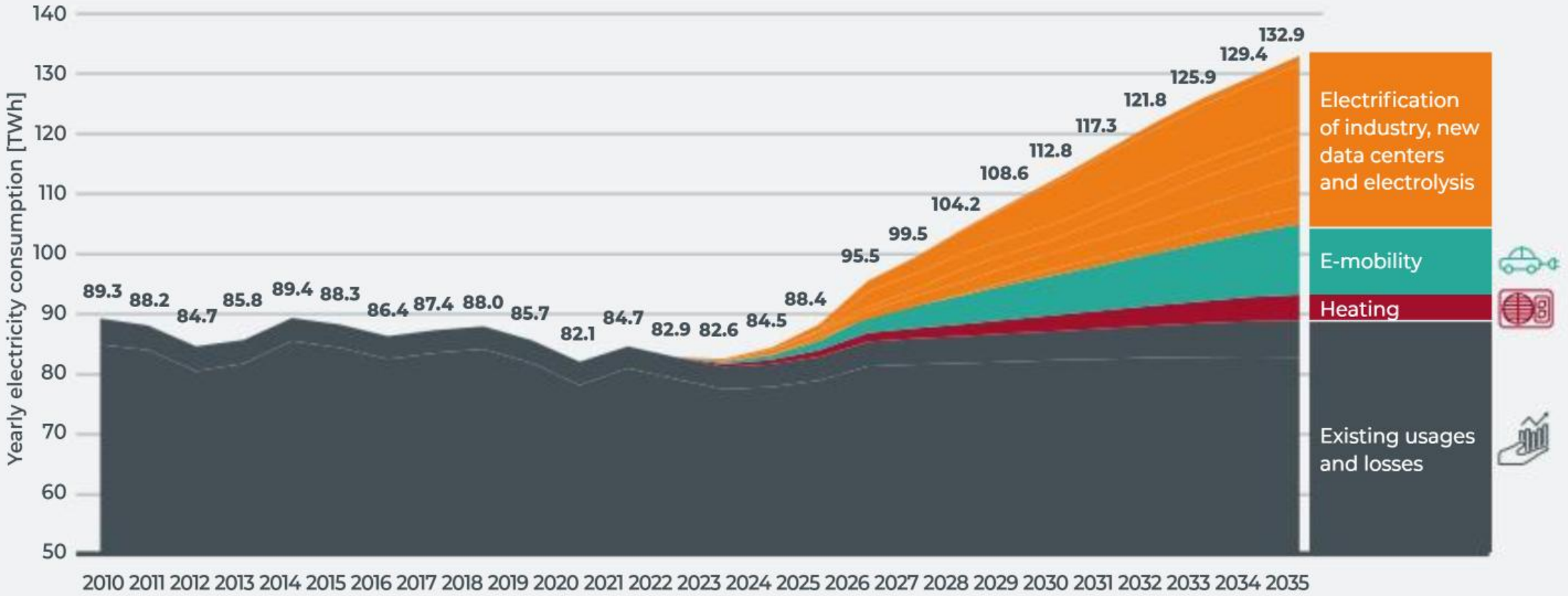
- Based on information from literature, consultations and expert views
- Consulted and updated every study based on latest available information
- Two main improvements on consumer flexibility
  - Integration of the new-large scale loads (electric boilers and ovens, as well as electric arc furnaces)
  - Improvement of end-user flex assumptions based on observations (e.g. EV charging)

Conventional Thermal CCGT, OCGT, large CHP, diesel and turbojets	**Dispatch constraints** (according start-up time)
Large-scale storage pumped-hydro, large batteries	**Energy reservoir Efficiency parameters**
Existing market response Existing processes	**Max activation duration**
Power-to-gas electrolysers	**Operational constraints**
New large-scale loads Electric arc furnaces, e-boilers and electric ovens	**Industry specific constraints**
Nuclear power Doel & Tihange	**Subject to technical constraints**
Interconnections	**Mainly for the intra-day market** and later also via EU balancing energy platforms
RES and small CHP based on generation profiles	**Mainly 'Profiled'** (only large wind / solar power parks, biofuel and small CHP)
Decentralised end-user flexibility Heat pumps, electric vehicles and home batteries	**Comfort limitations Energy limits**

- ↑ Upward flexibility considered in the calculations
- ↓ Downward flexibility considered in the calculations
- ↓ Downward flexibility not considered in the calculations (additional flexibility which might be available under exceptional conditions)

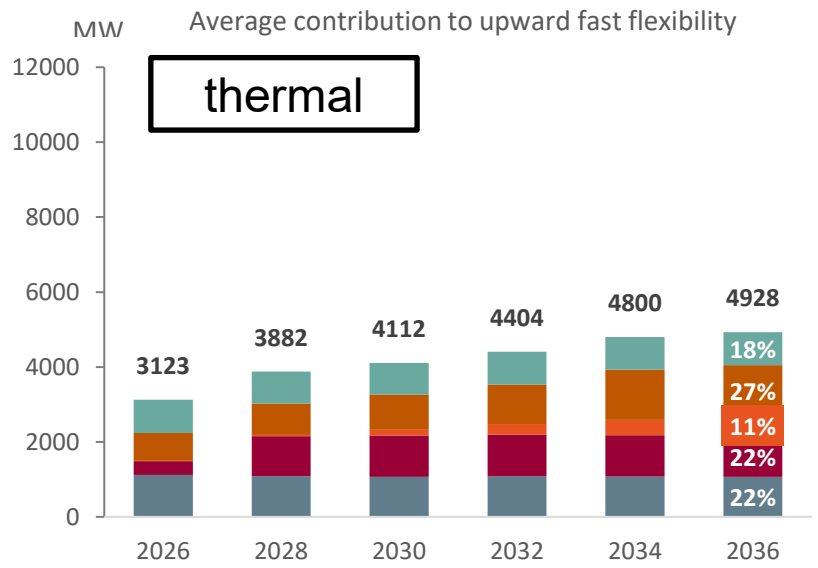
Electrification creates new sources of flexibility delivered by end-users

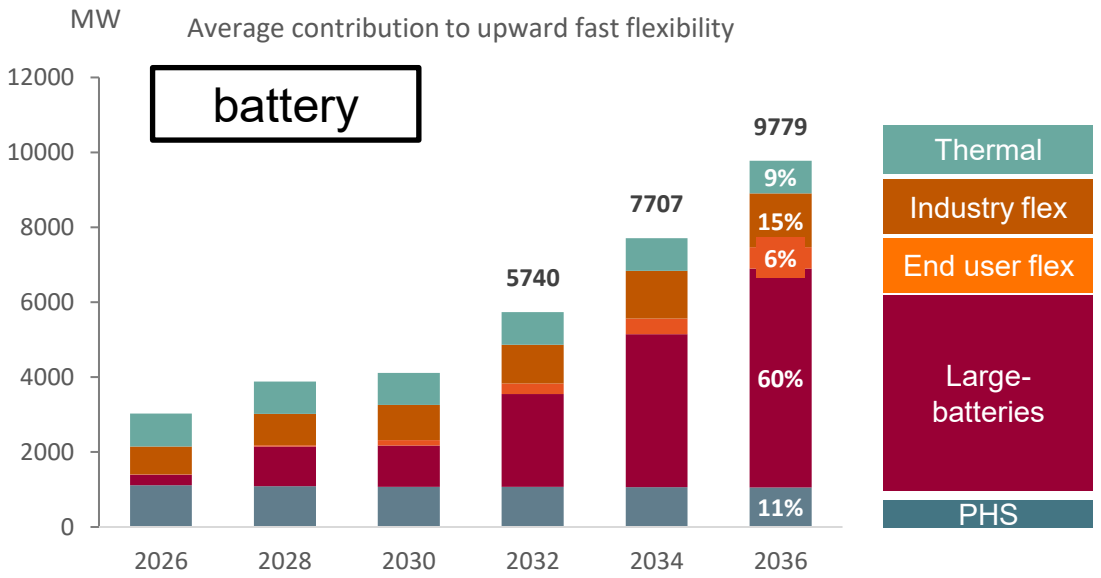
Figure 1: Changes in Belgium's electricity consumption between 2010 and 2035 (historical and estimated values)



Electrolysers and power-to-heat are an output of the economic dispatch model
Source: Elia (2023), 'Adequacy and Flexibility Study for Belgium (2024-2034)'

Availability of flexibility towards 2036 is largely driven by batteries and consumer flexibility





End-user flexibility is pre-dominantly driven by electric vehicles and home batteries

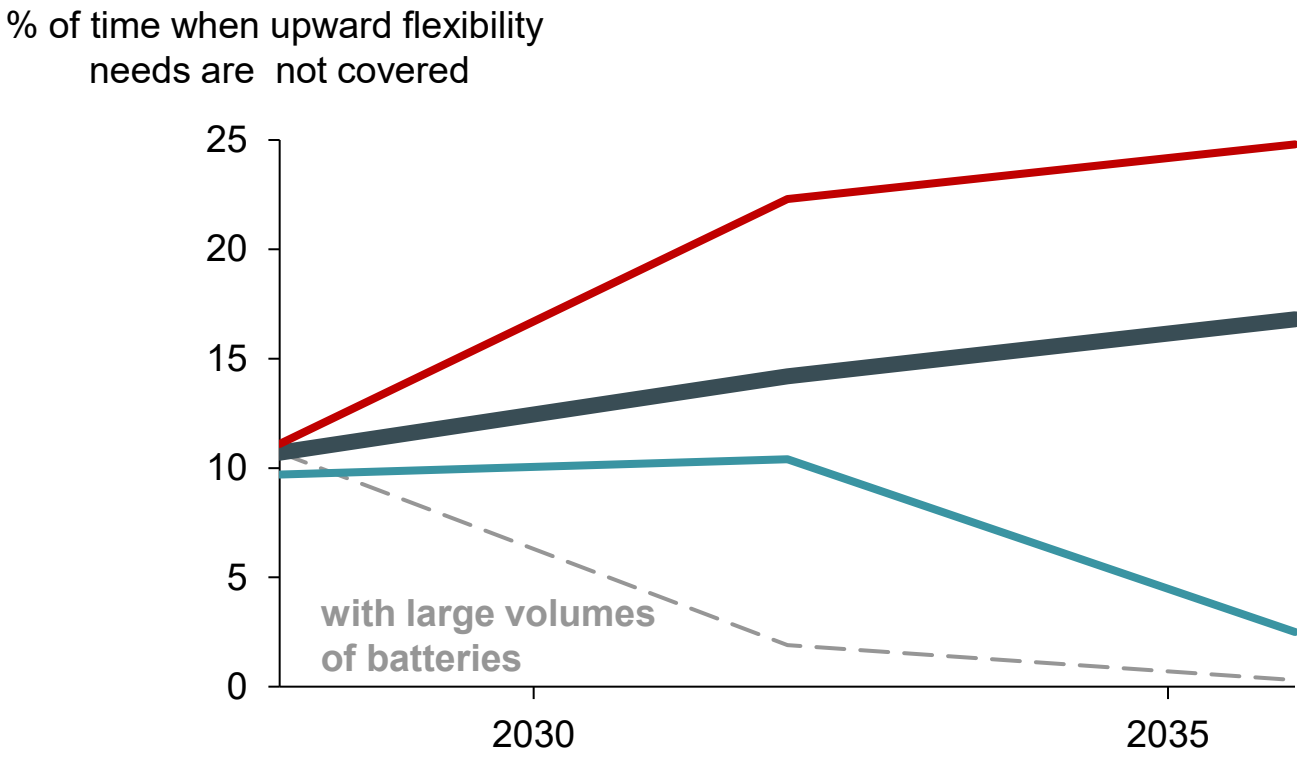
Large-scale loads are pre-dominantly driven by e-boilers, e-ovens and electric arc furnaces

- This contribution faces hour to hour variations according to the economic dispatch
- The availability of short-term flexibility depends largely on the scenario (thermal versus battery-driven GAP)
- A high uptake of large-scale battery storage is expected to dominate liquidity in short-term markets
- Consumer flexibility complements battery storage and provides a substantial contribution, particularly in thermal scenarios

Sensitivities show that future contributions of batteries and end-user flexibility are found to increase coverage levels

- *Higher coverage levels reduces the needs for reserve capacity or additional measures to be taken by Elia*
- *The contribution of end-user flexibility needs to be unlocked to cover flexibility needs*

Capacity is installed, but not always available when needed



Low end-user flexibility unlocked

% of time when fast flex needs not covered*

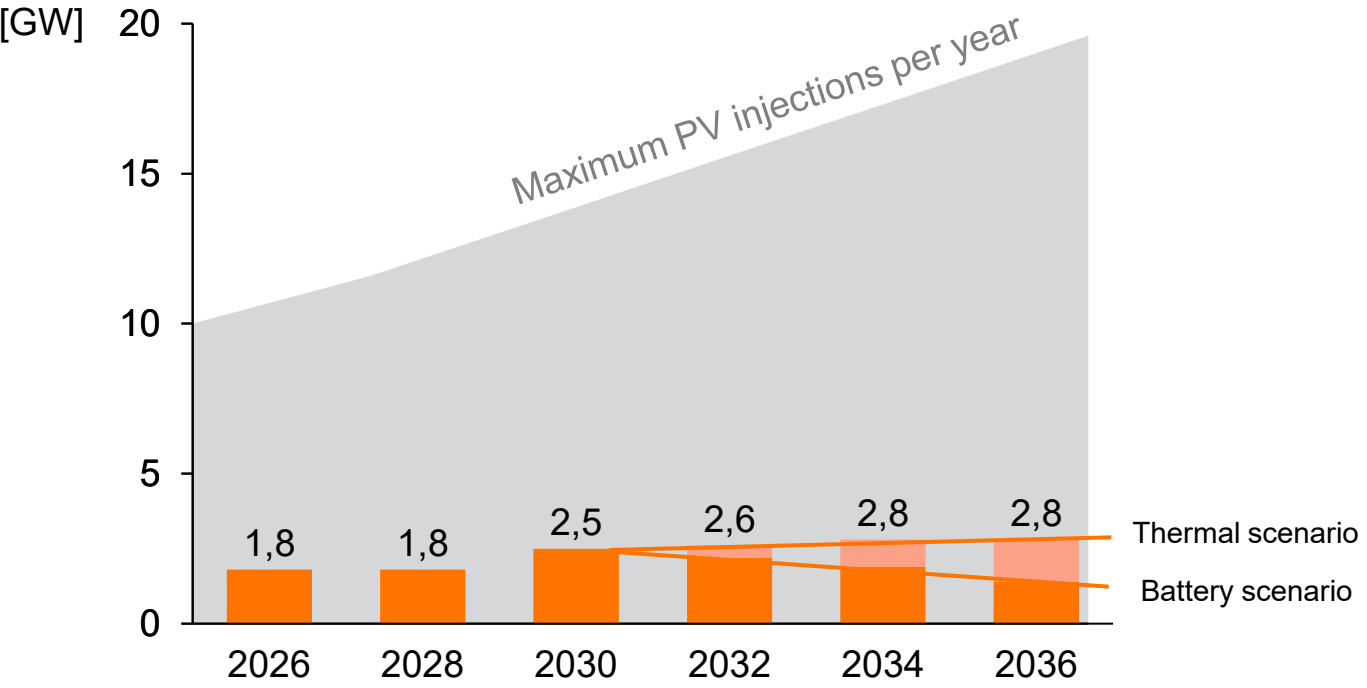
High end-user flexibility unlocked

* in the EU-BASE-CC 'thermal gap filling' scenario

Additional flexibility from decentralized assets is required to keep the system in balance

- From **300** (in 2026) towards **300-600 hours** (in 2036) with risks of not covering all flexibility needs*
- Additional **1,8 GW** (in 2026) to **2,5 GW** (in 2030) of consumer flexibility and decentral PV** flexibility is needed to react in the market to manage system imbalances

Decentralized PV or consumer flexibility to be unlocked during incompressibility risks periods to keep the system in balance at all times



Results represent minimum flexibility needs under scenario including all flexibility options of batteries, end-user flexibility, large-scale RES, cross-border flex under the Current Commitments scenario while accounting the contributions of decentralized wind power

Measures are

1. Intra-day and day-ahead is raising awareness among r situations are at risk
2. Remaining issues need to consumer flexibility and de market

** depending on the scenario*

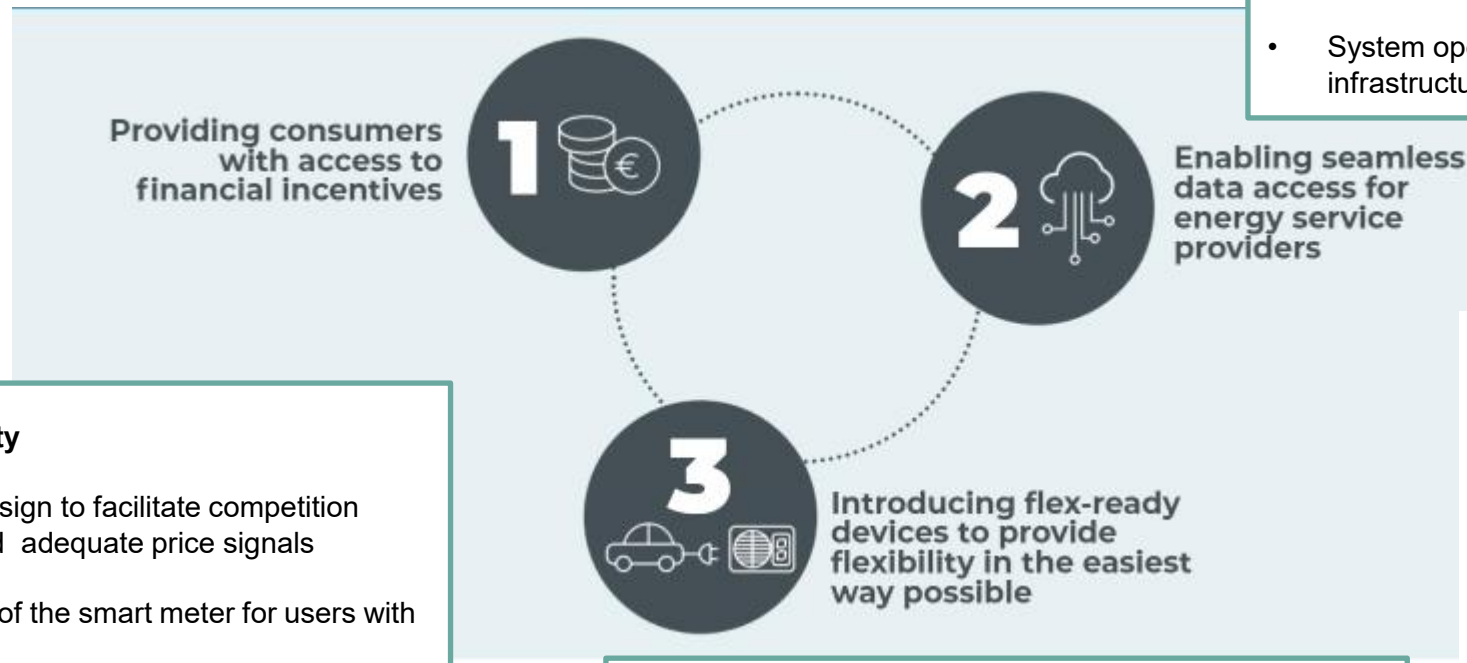
*** equivalent effective capacity*

**** interpolated*

Solving the barriers for flexibility

Ensure access of commercial parties to the data they need to deliver services to the consumer (secure, privacy, access rights and interoperable)

- Implementation of EU Regulation
- System operators can take lead on building the data infrastructure



Unlock value of flexibility

- Enhanced market design to facilitate competition behind the meter and adequate price signals
- Prioritize the roll-out of the smart meter for users with flexible assets
- Expanding consumer knowledge about energy
- Expanding access to time-based energy contracts and smart grid fees

Ensure that assets have minimum technical requirements to provide flexibility

- Develop Europe wide standard definitions on flex readiness to provide transparency to consumers
- Ensure future proof and interoperable communication standards

The Power of Flex

Studies (elia.be)

Structure

- Introduction
- Challenges of the energy transition
- Balancing a renewable energy system
- Flexibility needs of the system
- **Summer outlook**

Problem setting | Incompressibility Risks

The concern about incompressibility follows a combination of two separate challenges:

A. **Ability of the market to manage well 'predicted' situations of high renewable generation. It is related to the ability of market parties to maintain a balanced portfolio during high renewable energy conditions**

Perfect foresight
(Issue A)

B. **Ability of the system to maintain sufficient flexibility to manage unexpected outages or forecasting error. It is related to available downward flexibility in the system**

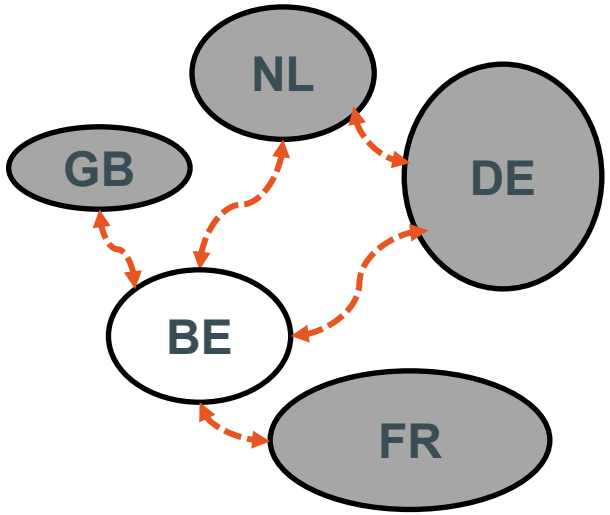
Deviations to perfect foresight
(Issue B)

Methodology | Summer outlook 2026 (versus 2025)

Summer outlook 2025

European power system simulation

Without neighboring countries influence



Incompressibility Risk KPI Definition:

Probabilistic # Hours Export Need* > Fixed Export Limit

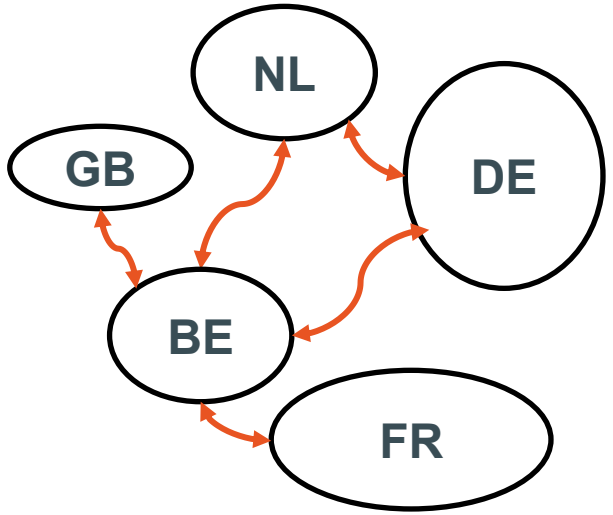
1. **No analysis on influence neighboring countries**
   Export capabilities to neighboring countries not taken into account in the analysis

2. **Limited analysis of type B issue, focusing on solar forecast uncertainty evolution**

Summer outlook 2026

European power system simulation

With neighboring countries influence



Incompressibility Risk KPI Definition:

Probabilistic # Hours Volume/Hours at risk after Export

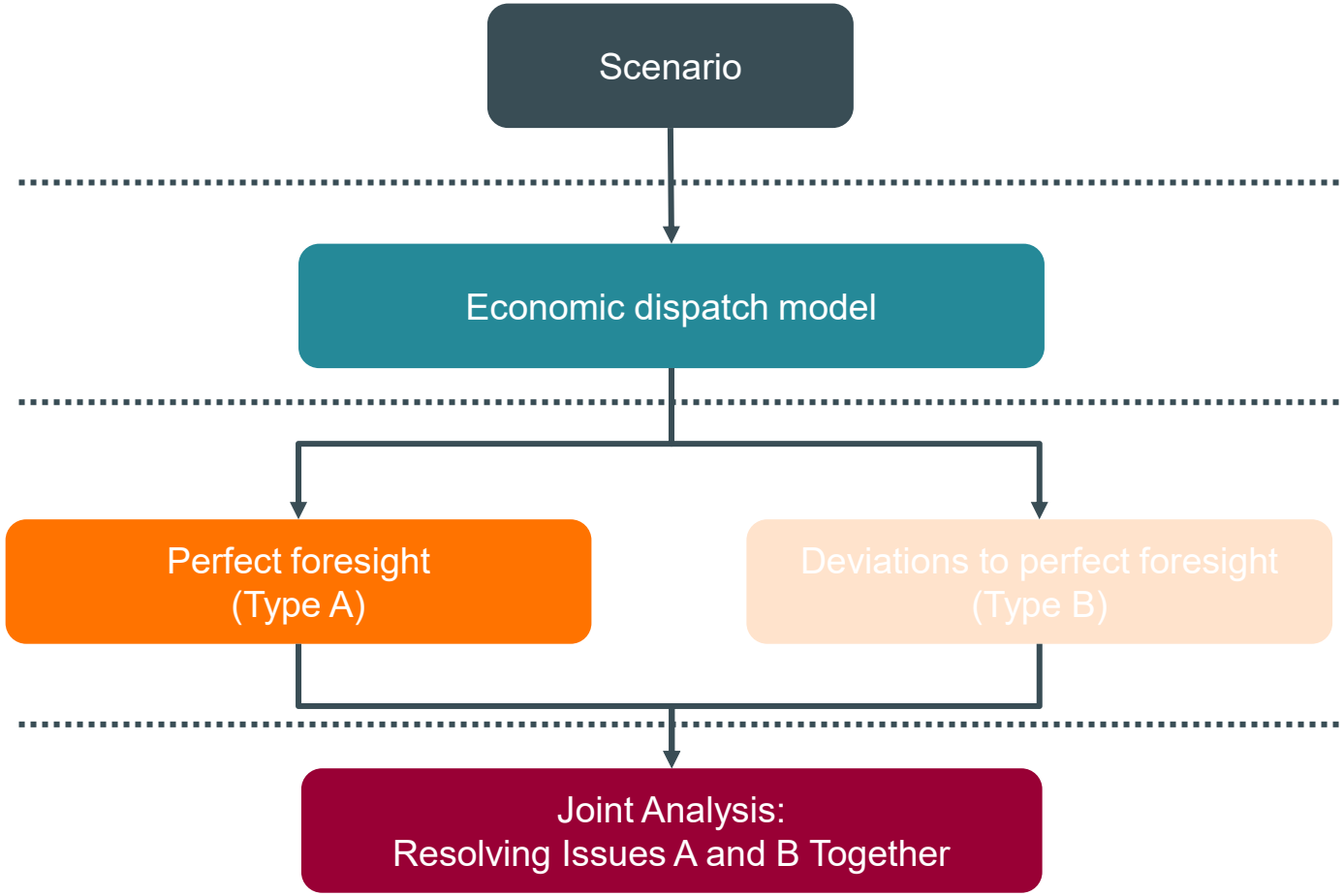
1. **Improved Incompressibility Indication**
   Correlation of incompressibility issues with neighboring countries

2. **First quantification of the type B issue, highlighting short-term flexibility needs**

Summary assets & grid evolutions compared to SO 2025

Higher Load	Regular increase of total load in BE compared to 2025
Less Nuclear	Decrease of ~ 2GW nuclear in BE from April onwards
More Solar	Increase of ~1 GW installed Panel PV capacity compared to SO25 scenario
More Onshore	Increase of ~ 277 MW installed Onshore capacity compared to SO25 scenario
More Batteries	Increase of ~ 500 MW installed battery capacity compared to SO25 scenario
More Hydro	Higher availability of hydro pump stations in BE compared to 2025

Summer outlook | type A – perfect foresight



Quantification of a scenario for the electricity system for the next summer across Europe.

Hourly economic dispatch model across numerous 'Monte Carlo' years, using Antares

A. Ability of the market to manage well 'predicted' situations of high renewable generation. It is related to the ability of market parties to maintain a balanced portfolio during high renewable energy conditions

B. Ability of the system to maintain sufficient flexibility to manage with unexpected outages or forecasting error. It is related to available downward flexibility in the system

Consolidation of analyses conducted on a perfect foresight model and on deviations due to unexpected outages or forecasting error.

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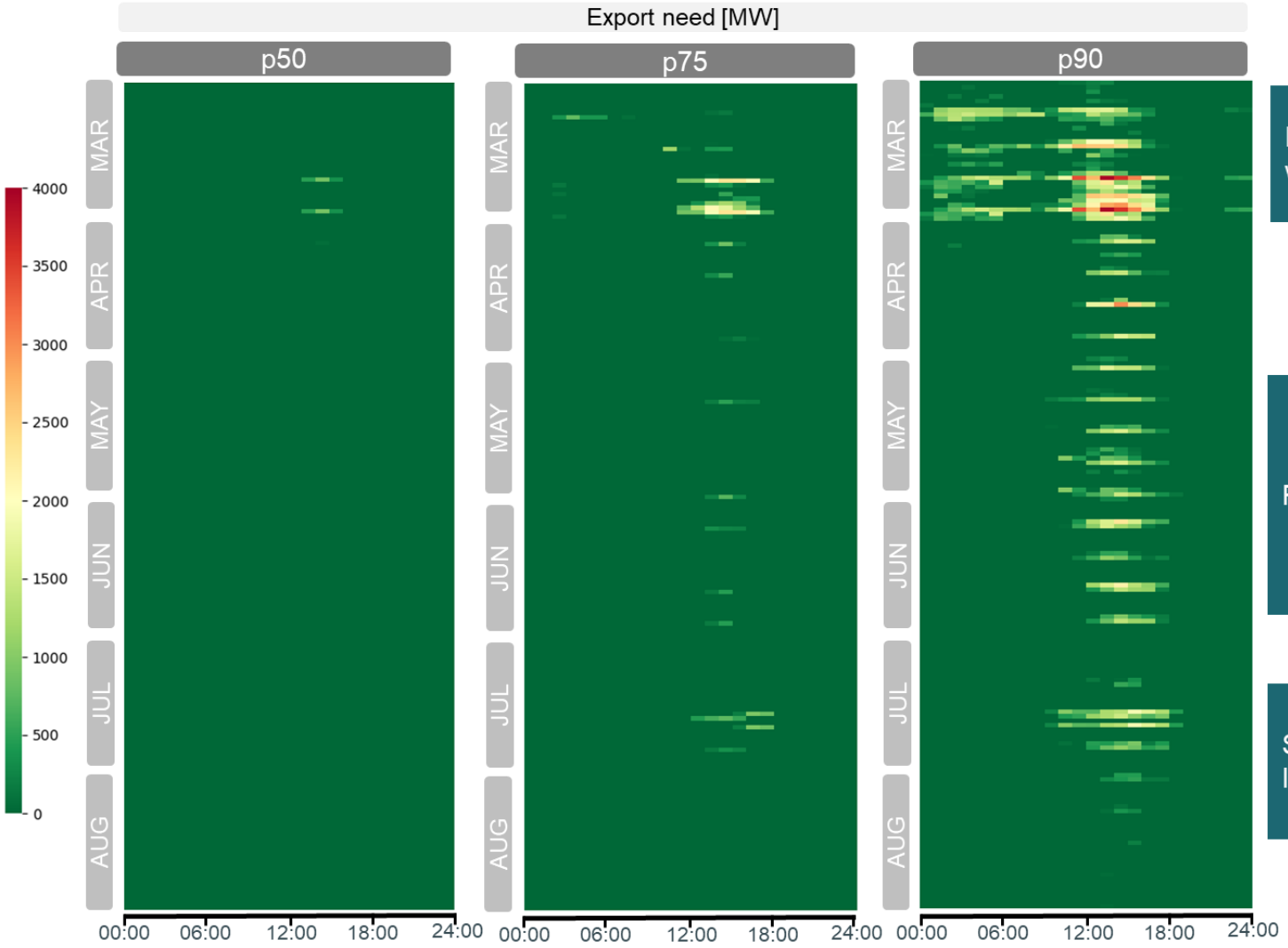
Summer outlook | type A – perfect foresight

Three main episodes of high export needs are expected



Nuclear running and higher wind production

RES production peak

Summer holidays leading to lower demand

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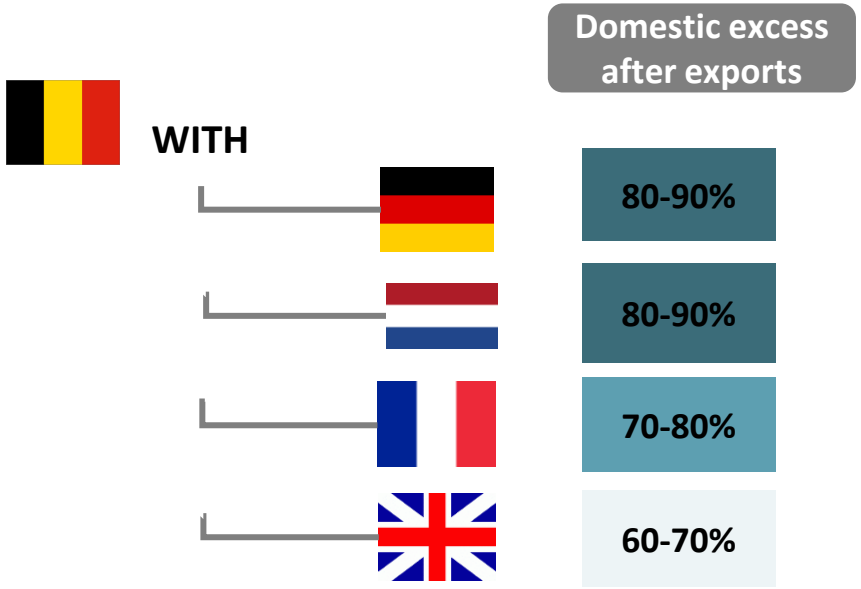
Summer outlook | type A – perfect foresight

When Belgium experiences domestic excess after export, neighbouring countries often face similar conditions

Probability that a neighbouring country experiences domestic excess after export at the same time as BE*



- The correlation of Germany and the Netherlands is the highest, mainly due to high correlation of RES generation**
- France has a lower correlation due to lower correlation of RES generation with BE**
- GB correlation is even lower but is expected to increase over the coming years due to upcoming wind projects in GB

*Range of probabilities across climate years

**Analysis done in AdeqFlex'25

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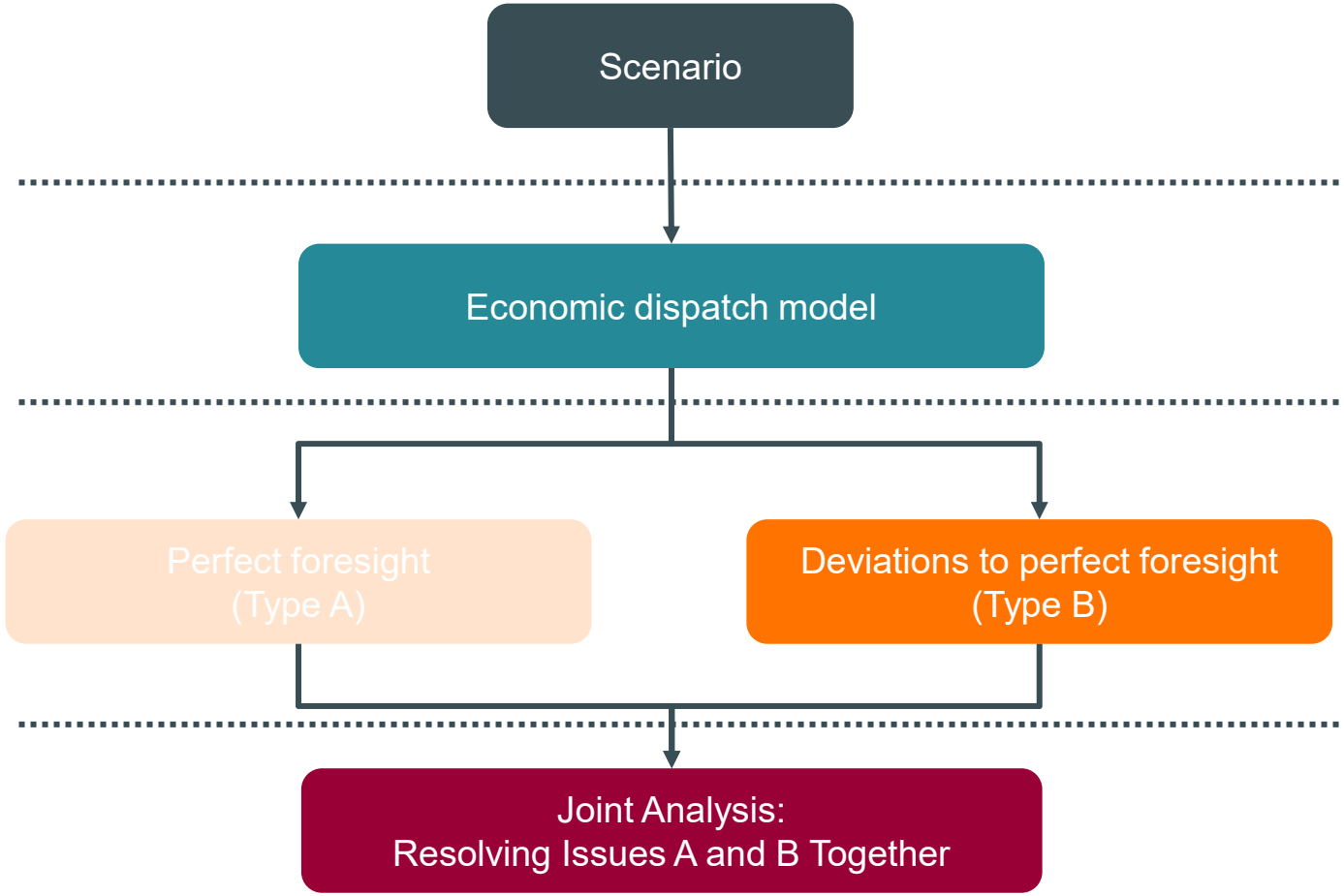
Summer outlook | type A – perfect foresight

The situation is less tense than previous summer

1. **Absence of nuclear generation** as from April
2. **Flexibility and storage** reduce the occurrence and the severity of the excess episodes
3. **Exports help mitigate excess generation** though to a smaller degree, as **neighboring countries** often face **domestic excess** simultaneously with Belgium
4. **Episodes of domestic excess after available RES curtailment** generally happen during the weekend, around noon
5. **Highest domestic excess tend to shift to Spring** while before they occurred mainly during summer

Summer outlook | type B – Deviations to perfect foresight



Quantification of a scenario for the electricity system for the next summer across Europe.

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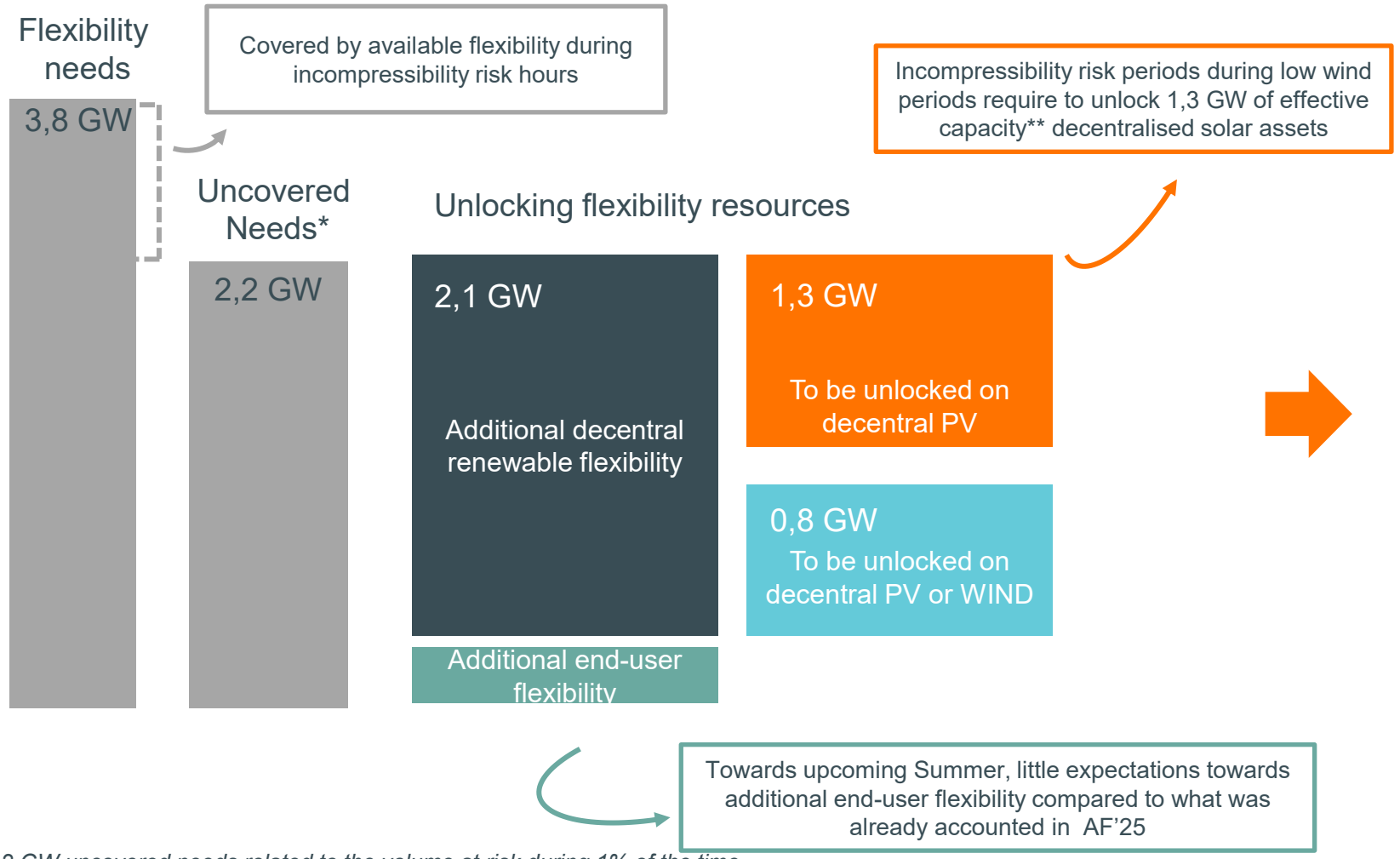
Consolidation of analyses conducted on a perfect foresight model and on deviations due to unexpected outages or forecasting error.

Summer outlook | type B – Deviations to perfect foresight

Uncovered volumes at risk shows that 1,3 GW needs to be unlocked on decentral solar assets from day-ahead to real time



Although there will be some additional market reaction on small RES assets, **Elia structurally depends on extreme 'last resort' measures to close this remaining gap:**

- (emergency) support from neighboring TSOs,
- activate redispatch bids for balancing and if needed,
- activate TSO and DSO technical measures

* *2.2 GW uncovered needs related to the volume at risk during 1% of the time*
** *Generating power during incompressibility*

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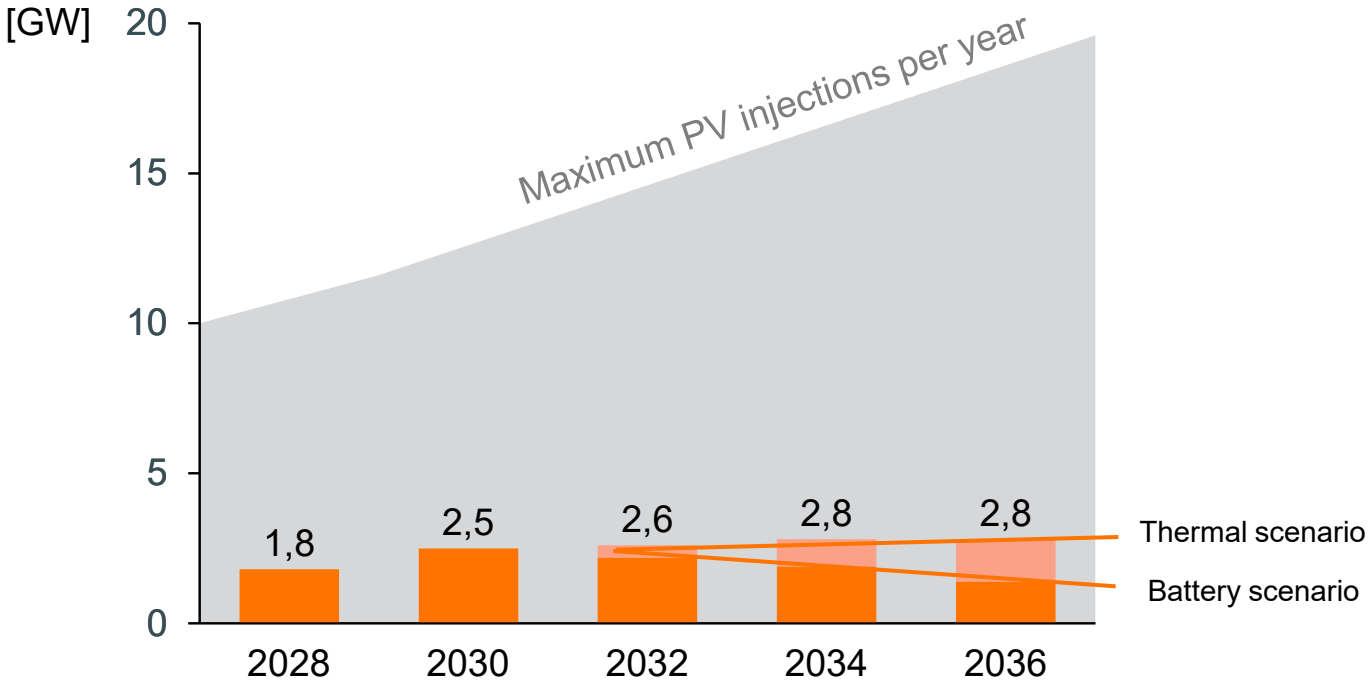
Summer outlook | type B – Deviations to perfect foresight

Uncovered flexibility shortages are expected to become worse in the coming years with increasing renewables (mainly PV) and the return of nuclear during spring & summer (as from 2029)

Adeqflex'25 : Decentralized PV or consumer flexibility to be unlocked during incompressibility risks periods



Increasing renewables (mainly PV) and nuclear as of summer 2029 increase the **need of unlocking additional decentralized capacity** during incompressibility risk periods

Results represent minimum flexibility needs under scenario including all flexibility options of batteries, end-user flexibility, large-scale RES, cross-border flex under the Adeqflex'25 Current Commitments scenario while accounting the contributions of decentralized wind power

Summer outlook: type B – Deviations to perfect foresight

Uncovered needs are covered through 'last resort' measures, additional flexibility must be unlocked

1. **Uncovered flexibility needs are observed**
   typically occurring during spring and summer, mostly during the weekends with low load, low wind and high-solar infeed

2. **Elia structurally depends on extreme 'last resort' measures to close the remaining gap**
   (emergency) support from neighbouring TSOs, redispatching bids for balancing and TSO/DSO technical measures

3. **Additional flexibility must be unlocked to cover 'worst case' solar-driven excess scenarios**

4. **Uncovered flexibility shortages are expected to become worse in the coming years**
   with increasing renewables (mainly PV) and the return of nuclear during summer (as from 2029)

Summer outlook: Conclusion

LESS TENSE SPRING & SUMMER '26
compared to '25

- Absence of **nuclear** generation
- **Flexibility and storage** contribute to excess management
- **Exports** support excess mitigation to a limited extent, due to regional correlation of events

Nevertheless…

ADDITIONAL FLEXIBILITY MUST BE UNLOCKED
to cover extreme events

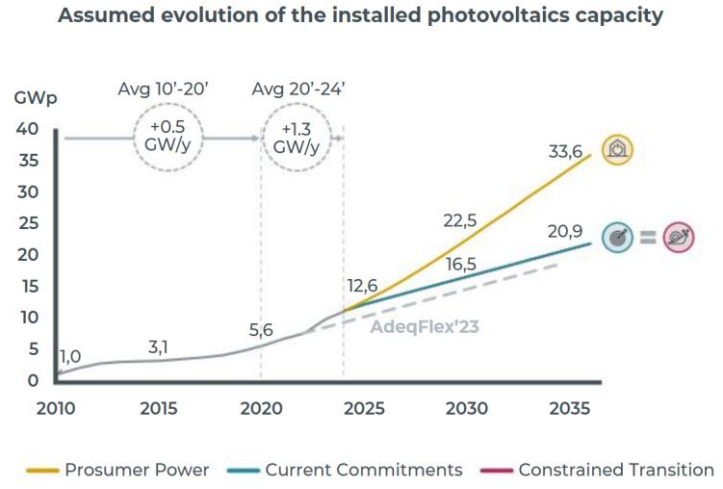
- **Uncovered flexibility is observed** during spring and summer weekends with low load, low wind and high-solar infeed
- **Elia structurally depends on extreme 'last resort' measures** to close the remaining gap: (emergency) support from neighbouring TSOs, redispatching bids for balancing and TSO/DSO technical measures
- **Additional flexibility must be unlocked to cover 'worst case'** solar-driven excess scenarios
- **Flexibility shortages expected to become worse** as renewables grow (mainly PV) and nuclear returns as from spring and summer 2029

CALL FOR ACTION

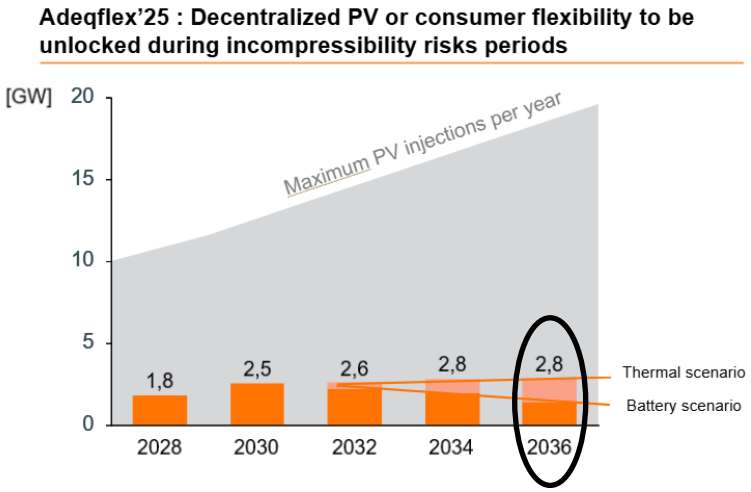
- Despite that Elia **welcomes the effort of market players** to respond to market conditions and new products being installed in the market, situations with **uncovered flexibility needs remain**.
- The existing **technical measures** on controllability of end-users and PV, need to be **anchored** in a solid legal framework and extended to smaller decentralised production units.
- Aligned with European legislation, Elia (together with the DSOs) will present this Summer an **update** of the **long-term Flexibility needs assessment** (BE) **towards 2030 and 2035.**

State of play | A growing gap between the need for flexibility and the number of distributed energy resources (DER) such as solar panels, heat pumps, electric vehicles, batteries, and others

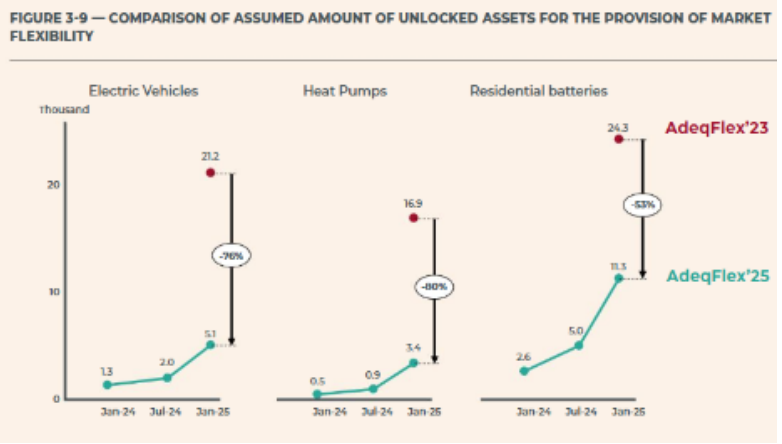
The production of renewable energy will exceed consumption in Belgium and the EU.



By 2036, at least 2.8 GW of flexibility needs to come from distributed energy resources (DER).



However, the figures for residential flexibility are much lower than expected



In line with our 'market first' principle, Elia is fully committed to facilitating **market-based flexibility.** At the same time, we cannot rely solely on the market to ensure grid security, and the **necessary and appropriate technical measures** must equally be put in place in a **timely manner.**

There is a need to **extend the current regulatory framework (e.g., a curtailment plan for production**), so that in exceptional circumstances, it becomes possible to technically control DER units below 1 MW